LAPORAN PENGERJAAN UJI KOMPETENSI KEJURUAN
SMK MAHARDHIKA BATUJAJAR
TAHUN PELAJARAN 2025/2026

Disusun Untuk Memenuhi Sebagian Persyaratan Kelulusan Tahun Pelajaran 2025/2026

Disusun Oleh :

Andira Althaf Aryaga **NIS 2324.10.350**

PROGRAM STUDI KEAHLIAN REKAYASA PERANGKAT LUNAK (RPL) SMK MAHARDHIKA BATUJAJAR KAB. BANDUNG BARAT
2025/2026

LAPORAN PENGERJAAN UJI KOMPETENSI KEJURUAN
SMK MAHARDHIKA BATUJAJAR
TAHUN PELAJARAN 2023-2024

KOMPETENSI KEAHLIAN : Rekayasa Perangkat Lunak
KELAS : XII RPL 1
NAMA LENGKAP : Andira Althaf Aryaga
NIS : 2324.10.350
PAKET SOAL : P2
JENIS APLIKASI : Web
ALOKASI WAKTU : 24 jam

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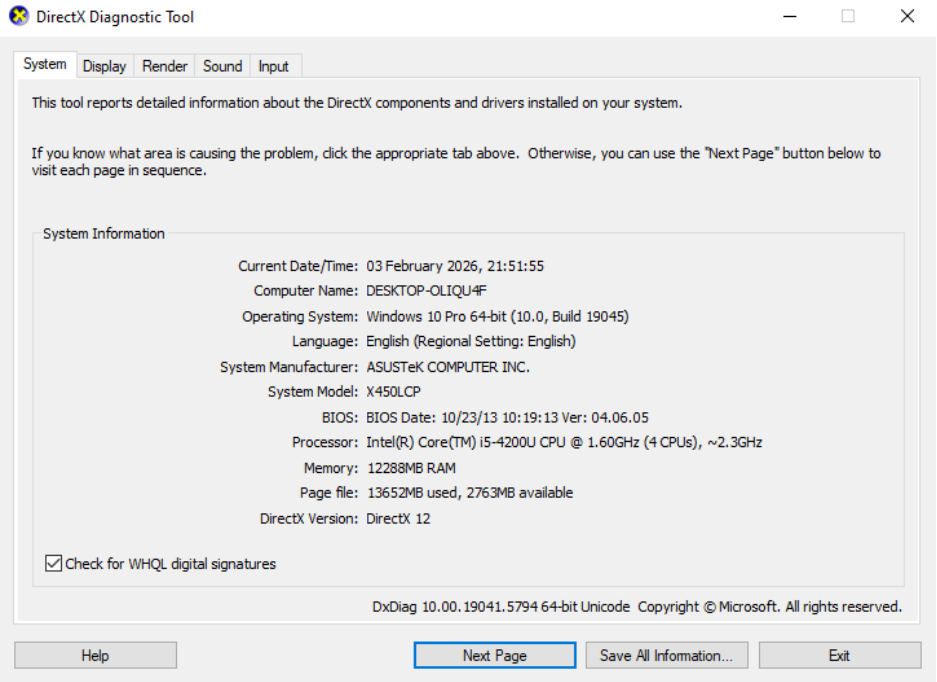
1. PENJADWALAN KEGIATAN

KEGIATAN	ALOKASI WAKTU														
	JANUARI	FEBRUARI				MARET					APRIL				
	28 - 29	3 - 5	10 - 12	17 - 19	24 - 26	3 - 5	10 - 12	17 - 19	24 - 26	31	1 - 2	7 - 9	14 - 16	21 - 23	28 - 30
Analisis Soal															
Persiapan Hardware															
Persiapan Software															
Pembuatan Gantt Chart															
Perancangan Sistem															
Perancangan Database															
Perancangan Desain UI															
Implementasi Database															
Implementasi Desain UI															
Implementasi Kode Program															
Pengujian Perangkat Lunak															
Pembuatan Dokumen User Manual															
Dokumentasi															
Finalisasi Dokumen dan Program															
Hosting															
Github															

2. PERSIAPAN PERANGKAT KERAS



Gambar 1.1 Spesifikasi Device yang digunakan

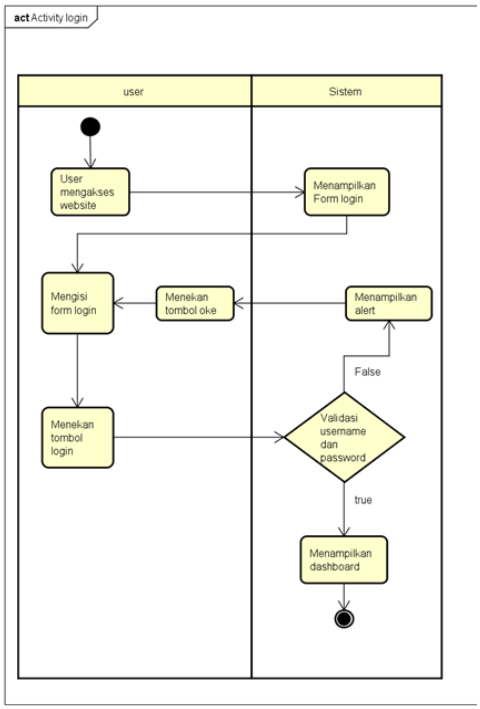
1	**RFID**	RFID merupakan sebuah perangkat teknologi nirkabel yang memiliki kemampuan untuk mengidentifikasi target atau objek tertentu, membaca dan memasukkan data melalui sinyal radio.	
2	**MOTOR SERVO**	Motor servo merupakan perangkat atau aktuator putar (motor) yang dirancang dengan sistem kontrol feedback loop tertutup (close loop), sehingga dapat memastikan dan menentukan posisi sudut dari poros output motor.	
3	**ESP32**	ESP32 adalah mikrokontroler yang mendukung dalam pengembangan Internet of Things karena kemampuan yang kuat dan harga yang terjangkau.	
4	**OLED**	OLED (Organic Light Emitting Diodes) adalah teknologi pemancar cahaya datar, yang dibuat dengan menempatkan serangkaian film tipis organik di antara dua konduktor. Ketika arus listrik dialirkan, cahaya terang akan dipancarkan.	

3. PERSIAPAN PERANGKAT LUNAK

NO	SOFTWARE	FUNGSI
1	Visual Studio Code	Software teks editor digunakan untuk mengembangkan berbagai jenis aplikasi termasuk aplikasi web.
2	Laragon	Merupakan media atau web server localhost yang bisa digunakan secara offline.
3	Chrome	Browser ini digunakan untuk menguji aplikasi web.
4	Astah Professional	Alat pemodelan visual yang digunakan untuk merancang dan mengembangkan diagram UML dan diagram alur kerja.
5	Balsamiq	Alat prototyping wireframing yang memungkinkan pengguna untuk membuat mockup antarmuka pengguna (UI).
6	Microsoft Word	Software pengolah pengolah data yang digunakan untuk membuat, mengedit, dan memformat dokumen teks seperti surat, laporan, dan makalah.
7	Github	GitHub adalah platform berbasis cloud dan situs jejaring sosial untuk pengembang perangkat lunak, yang digunakan untuk menyimpan, mengelola, dan berbagi kode program menggunakan sistem kontrol versi Git.
8	MQTT Exproler	MQTT Explorer adalah aplikasi desktop (GUI) sumber terbuka yang populer digunakan untuk memantau, men-debug, dan menguji broker MQTT secara visual.
9	HiveMQ	HiveMQ adalah broker pesan MQTT enterprise kelas atas yang dirancang untuk komunikasi machine-to-machine (M2M) dan Internet of Things (IoT) secara real-time, andal, dan aman.
10	ArduinoIDE	Arduino IDE (Integrated Development Environment) adalah perangkat lunak yang digunakan untuk memprogram dan mengunggah kode ke papan Arduino. Ini adalah alat inti yang digunakan oleh para pengembang, pemula, dan pecinta elektronika untuk mengembangkan berbagai macam proyek berbasis Arduino.

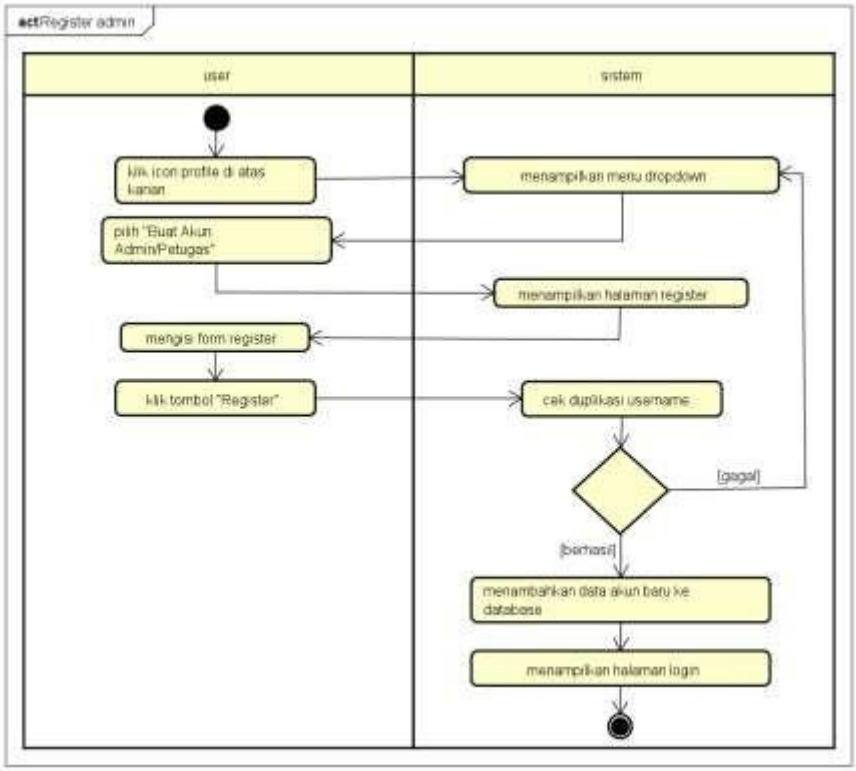
4. PERANCANGAN ALGORITMA APLIKASI

1) Login



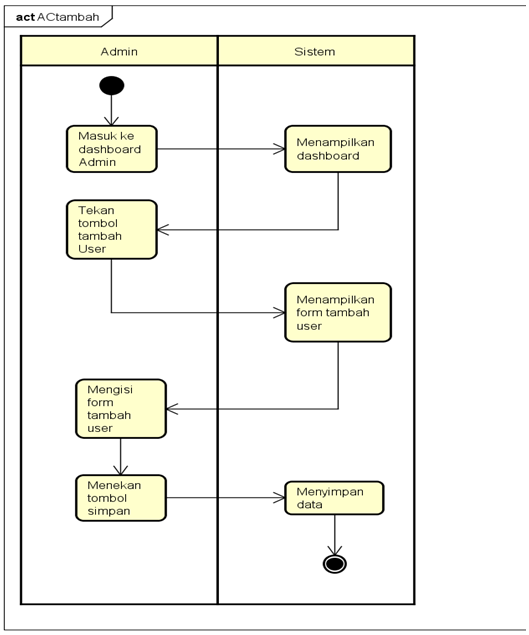
Gambar 1.2 Activity Diagram Registrasi Login

2) Logout



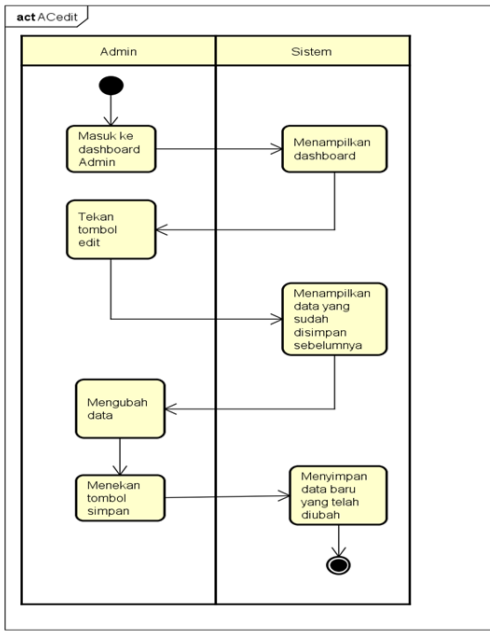
Gambar 1.3 Activity Diagram Logout

3) Tambah User



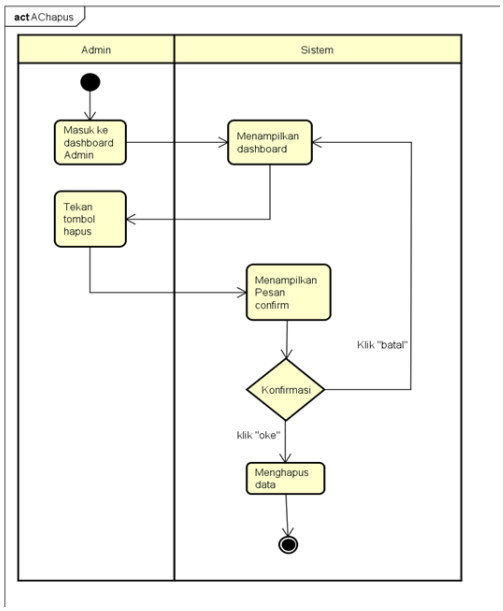
Gambar 1.4 Activity Diagram Tambah User

4) Edit User



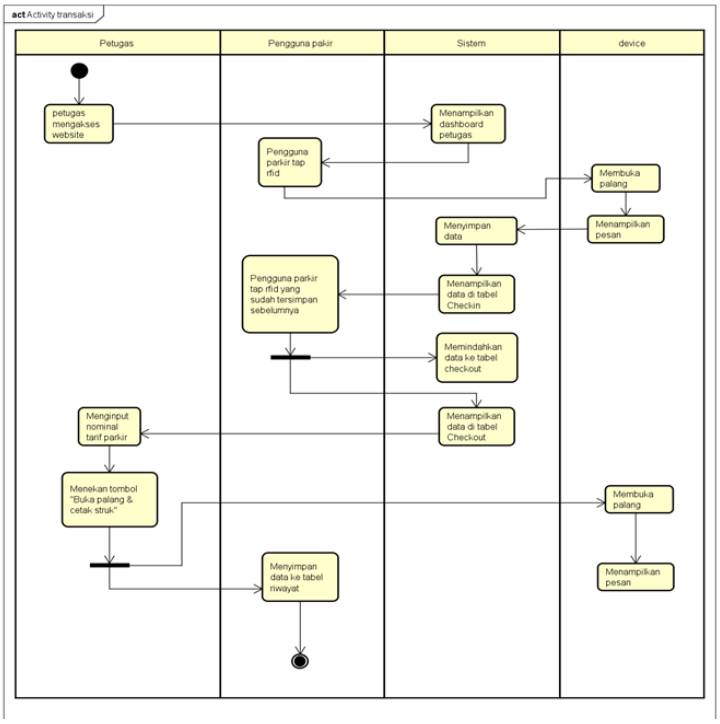
Gambar 1.5 Activity Diagram Edit User

5) Hapus User



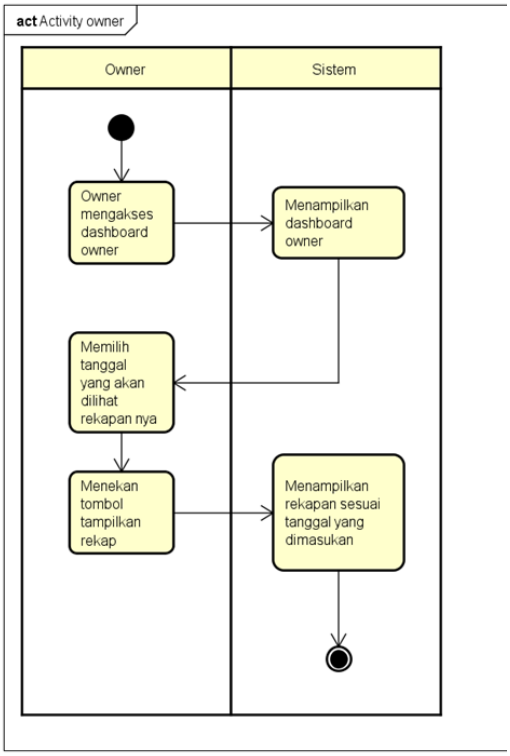
Gambar 1.6 Activity Diagram Hapus User

6) Transaksi Check In – Check Out



Gambar 1.7 Activity Diagram Transaksi Check In - Check Out

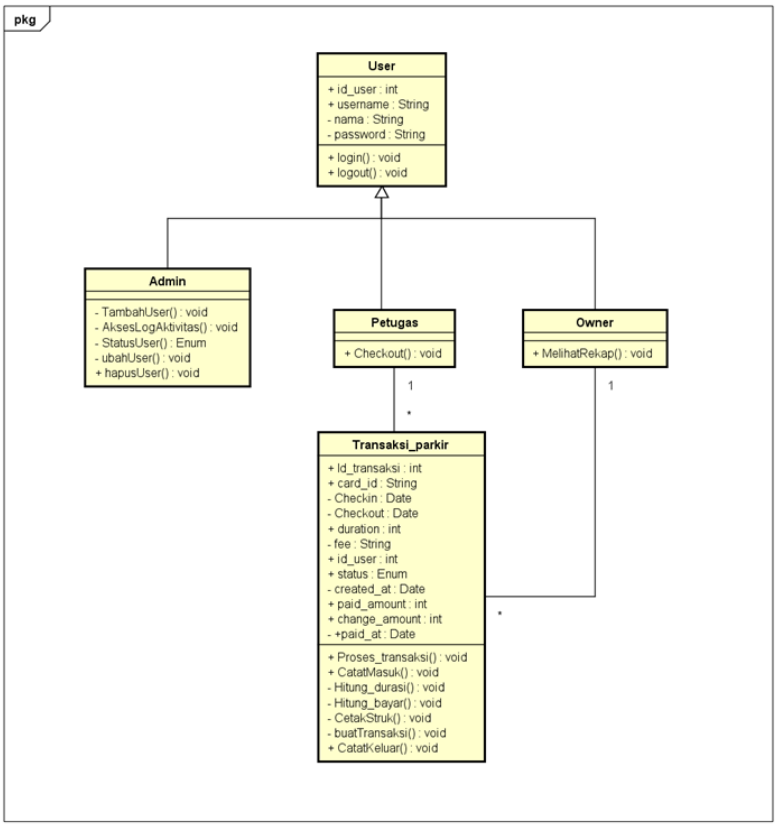
7) Owner



Gambar 1.8 Activity Diagram Owner

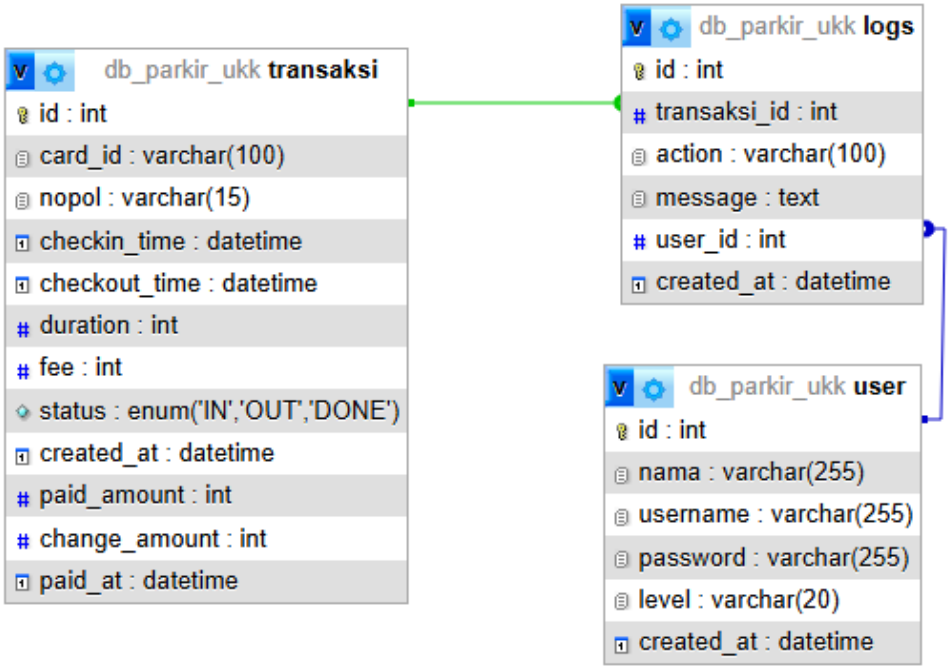
5. PERANCANGAN DESAIN BASIS DATA

1) Class Diagram



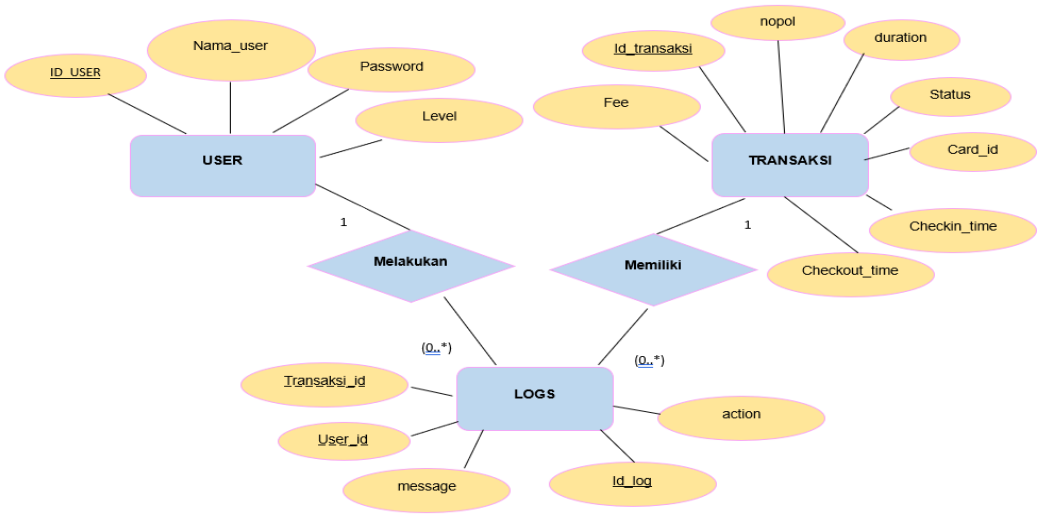
Gambar 1.9 Class Diagram

2) Table SQL Designer



Gambar 1.10 SQL Designer

3) ER Diagram



Gambar 1.11 Entity Relationship Diagram

6. DUMMY DATA

1) Users

id	Nama	Username	Password	Level	Created_at
15	Andira	1	1	Admin	2026-02-09 23:33:31
16	Althaf	2	2	Owner	2026-02-11 22:34:32
21	Aryaga	3	3	Petugas	2026-02-25 18:22:14

2) Transaksi

id	Card_id	No pol	Checkin_time	Checkout_time	duration	fee	status	Created_at	Paid_amount	Change_amount	Paid_at
21	TAG 1024	B 6141 QTZ	2026-03-10 08:58:55	2026-03-10 09:30:59	1	3000	DONE	2026-03-10 08:58:55	3000	0	2026-03-10 09:31:09
22	TAG 1117	B 1388 OSD	2026-03-10 08:59:55	2026-03-10 09:33:08	1	3000	DONE	2026-03-10 08:59:55	3000	0	2026-03-23 21:46:14
33	4112233	B 7337 WCT	2026-03-13 20:14:22	2026-03-23 21:45:24	242	726000	OUT	2026-03-13 20:14:22	*NULL*	*NULL*	*NULL*
35	1234	B 8362 THY	2026-03-23 21:50:47	*NULL*	*NULL*	*NULL*	IN	2026-03-23 21:50:47	*NULL*	*NULL*	*NULL*

3) Logs

Id	action	message	user	created_at
1	CHECKIN	eck-in untuk kartu TAG001	AABBBCCC	26-02-09 22:18:41
2	CHECKOUT	Check-out untuk kartu TAG001, Durasi: 0 menit Biaya: Rp0	AABBBCCC	26-02-09 22:19:02
3	CHECKIN	eck-in untuk kartu TAG002	AABBBCCC	26-02-09 22:20:45
4	CHECKOUT	Check-out untuk kartu TAG002, Durasi: 367 menit, Biaya: Rp14000	AABBBCCC	26-02-09 22:27:49

7. PEMBUATAN BASIS DATA (MySQL Code)

1) Users

#	Nama	Jenis	Penyortiran	Atribut	Tak Ternilai	Bawaan	Komentar	Ekstra	Tindakan
1	**id**	int(11)			Tidak	*Tidak ada*		AUTO_INCREMENT	Ubah Hapus Lainnya
2	**nama**	varchar(255)	utf8mb4_general_ci		Tidak	*Tidak ada*			Ubah Hapus Lainnya
3	**username**	varchar(255)	utf8mb4_general_ci		Tidak	*Tidak ada*			Ubah Hapus Lainnya
4	**password**	varchar(255)	utf8mb4_general_ci		Tidak	*Tidak ada*			Ubah Hapus Lainnya
5	**level**	varchar(20)	utf8mb4_general_ci		Tidak	*Tidak ada*			Ubah Hapus Lainnya
6	**created_at**	datetime			Tidak	current_timestamp()			Ubah Hapus Lainnya

Gambar 1.12 Tabel Users

2) Transaksi

#	Nama	Jenis	Penyortiran	Atribut	Tak Ternilai	Bawaan	Komentar	Ekstra	Tindakan
1	**id**	int(11)			Tidak	*Tidak ada*		AUTO_INCREMENT	Ubah Hapus Lainnya
2	**card_id**	varchar(100)	utf8mb4_general_ci		Tidak	*Tidak ada*			Ubah Hapus Lainnya
3	**nopol**	varchar(15)	utf8mb4_general_ci		Ya	*NULL*			Ubah Hapus Lainnya
4	**checkin_time**	datetime			Ya	*NULL*			Ubah Hapus Lainnya
5	**checkout_time**	datetime			Ya	*NULL*			Ubah Hapus Lainnya
6	**duration**	int(11)			Ya	*NULL*			Ubah Hapus Lainnya
7	**fee**	int(11)			Ya	*NULL*			Ubah Hapus Lainnya
8	**status**	enum('IN', 'OUT', 'DONE')	utf8mb4_general_ci		Tidak	IN			Ubah Hapus Lainnya
9	**created_at**	datetime			Ya	current_timestamp()			Ubah Hapus Lainnya
10	**paid_amount**	int(11)			Ya	*NULL*			Ubah Hapus Lainnya
11	**change_amount**	int(11)			Ya	*NULL*			Ubah Hapus Lainnya
12	**paid_at**	datetime			Ya	*NULL*			Ubah Hapus Lainnya

Gambar 1.13 Tabel Transaksi

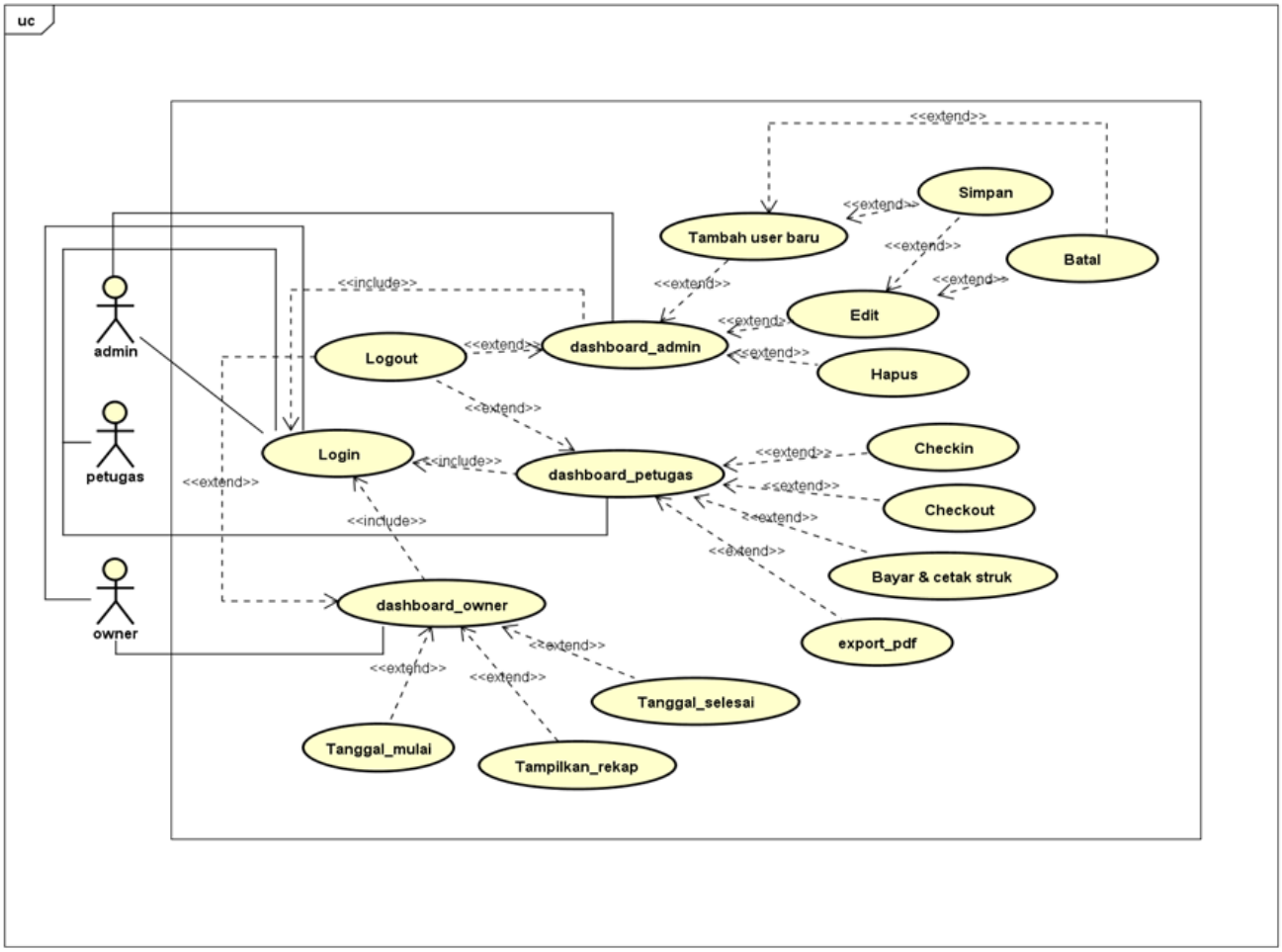
3) Logs

#	Nama	Jenis	Penyortiran	Atribut	Tak Ternilai	Bawaan	Komentar	Ekstra	Tindakan
1	**id**	int(11)			Tidak	*Tidak ada*		AUTO_INCREMENT	Ubah Hapus Lainnya
2	**transaksi_id**	int(11)			Ya	*NULL*			Ubah Hapus Lainnya
3	**action**	varchar(100)	utf8mb4_general_ci		Tidak	*Tidak ada*			Ubah Hapus Lainnya
4	**message**	text	utf8mb4_general_ci		Ya	*NULL*			Ubah Hapus Lainnya
5	**user_id**	int(11)			Ya	*NULL*			Ubah Hapus Lainnya
6	**created_at**	datetime			Ya	current_timestamp()			Ubah Hapus Lainnya

Gambar 1.14 Tabel Logs

8. PERANCANGAN TAMPILAN ANTARMUKA (UI) WEB

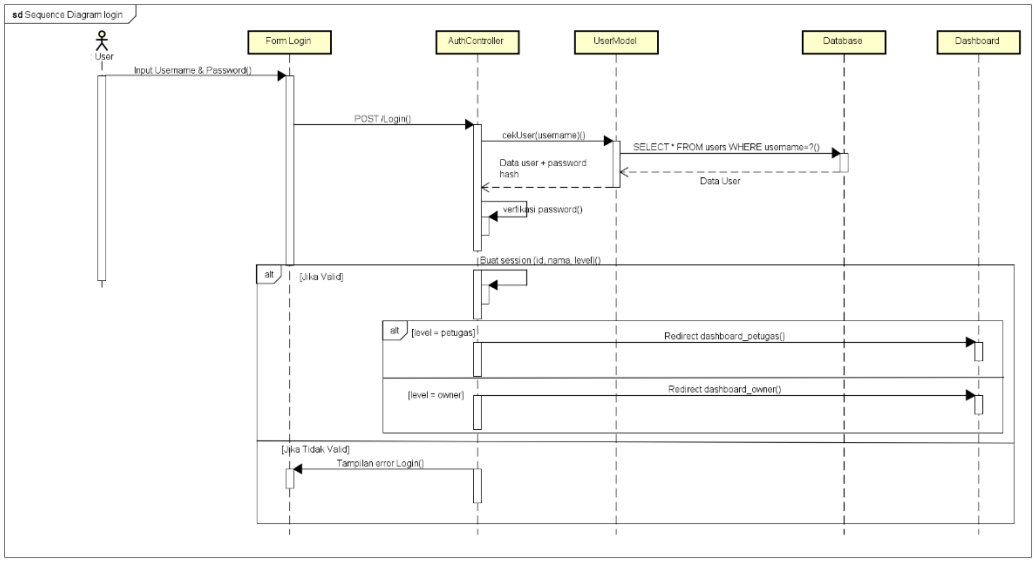
1. UseCase Diagram



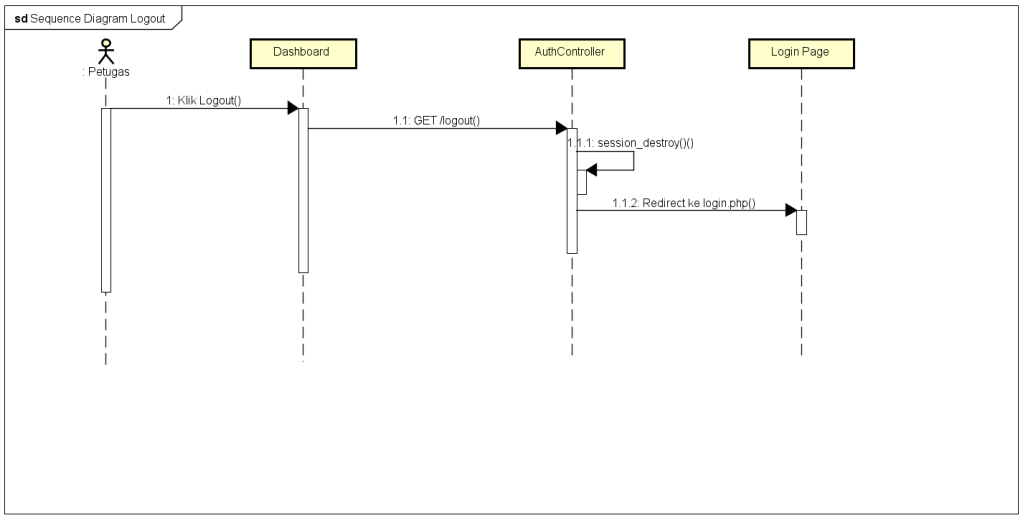
Gambar 1.17 UseCase Diagram

2. Sequence Diagram

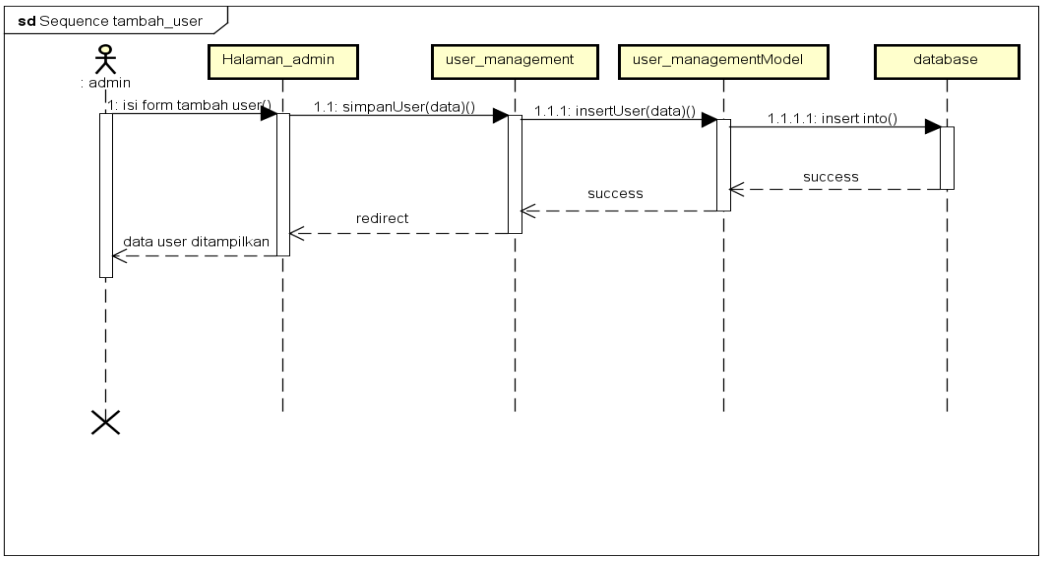
1) Login



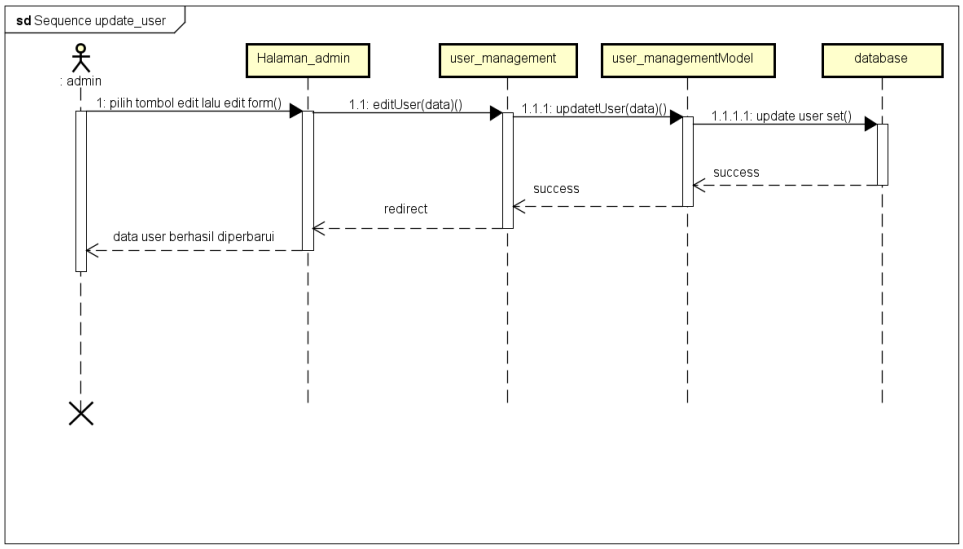
2) Logout



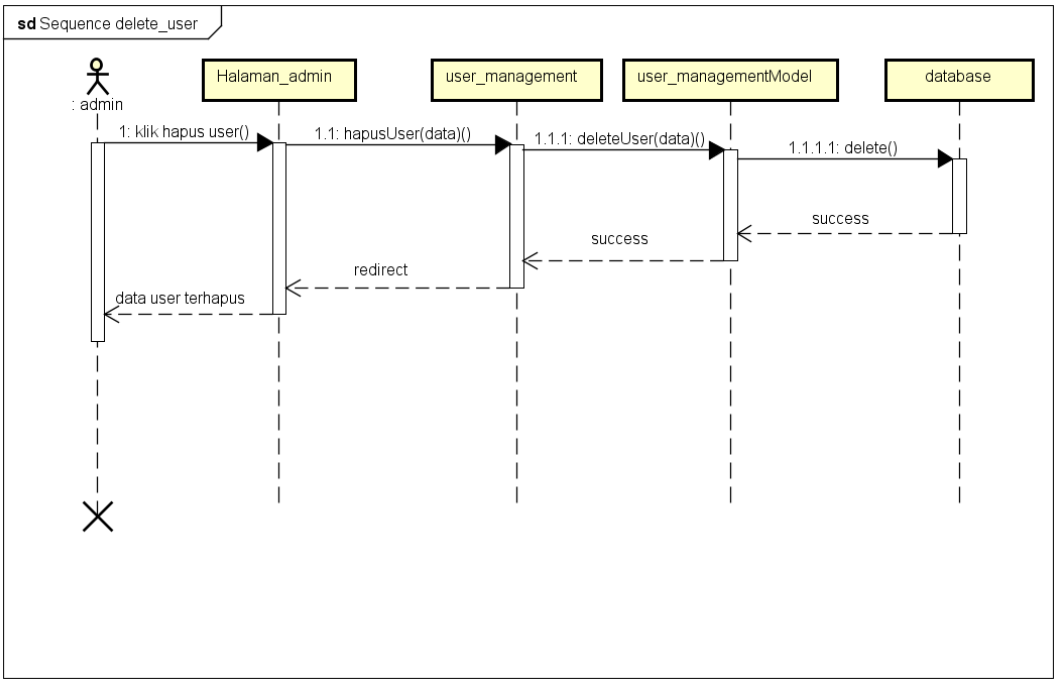
3) Tambah



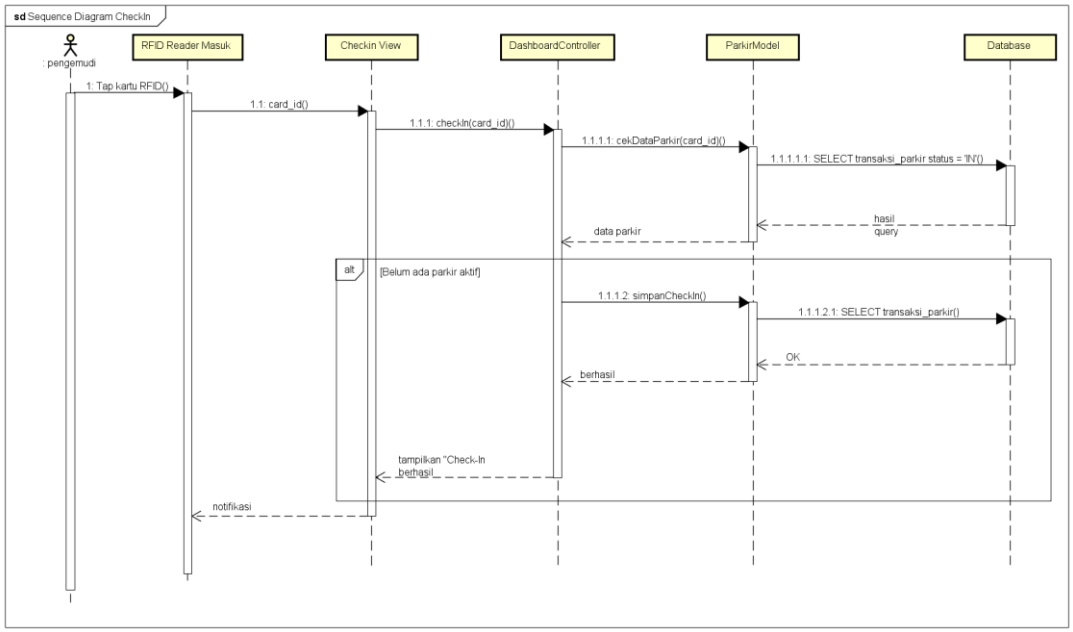
4) Edit



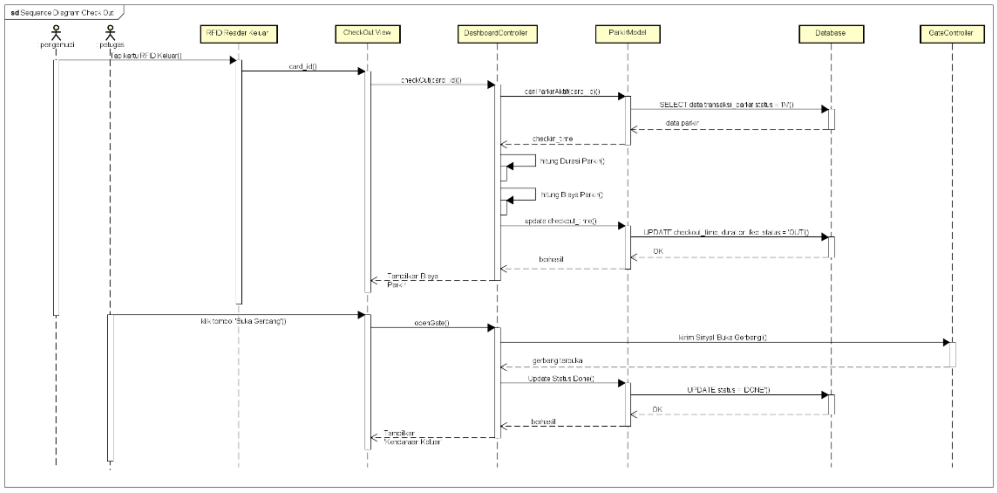
5) Hapus



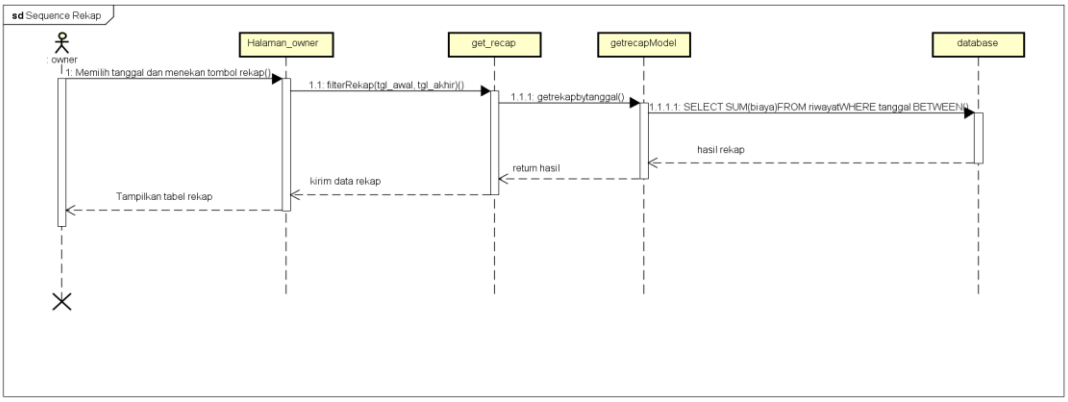
6) Check In



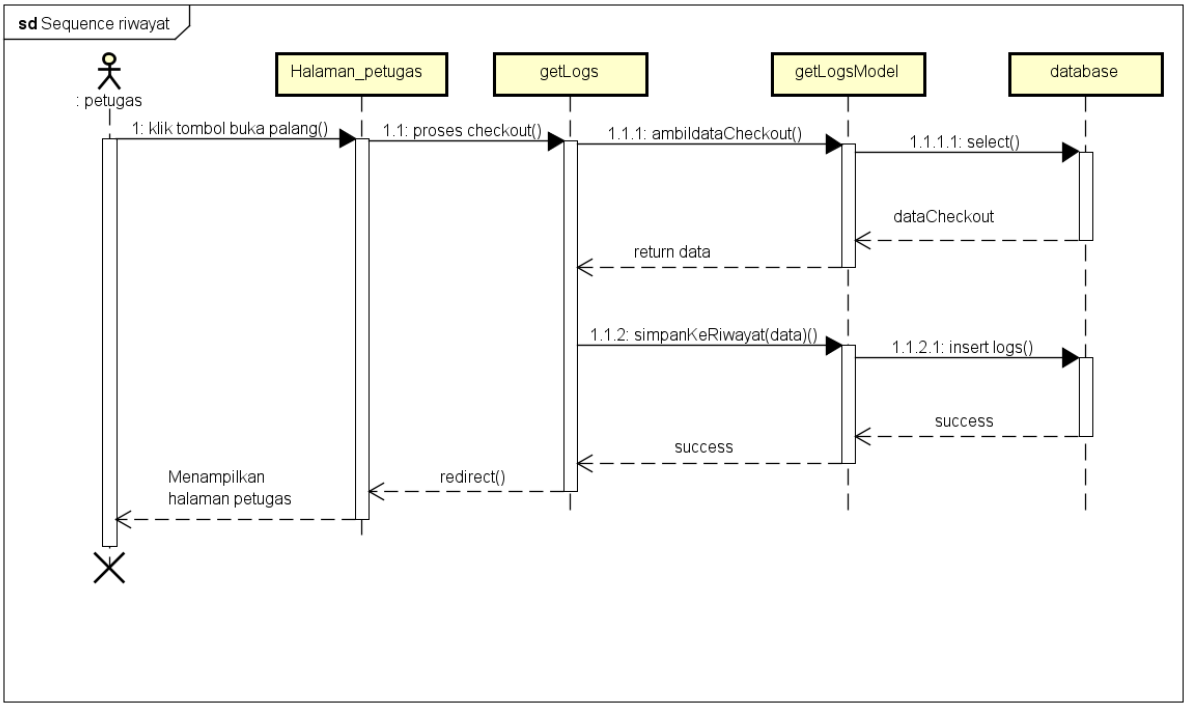
7) Check Out



8) Rekap



9) History



9. Skenario Use Case

Login

Name : Login
IDE : UC001
Description : Masuk menggunakan akun yang sudah terdaftar
Actor : User

Main Course	
Aksi Aktor	**Reaksi Sistem**
1. Masuk ke halaman website	
2. Memasukkan username dan password	
3. Klik tombol “login”	
	4. Menampilkan halaman dashboard

Tambah user

Name : Tambah
IDE : UC002
Description : Menambahkan/Mendaftarkan user baru
Actor : Admin

Main Course	
Aksi Aktor	**Reaksi Sistem**
1. Masuk halaman admin	
	2. Menampilkan halaman admin
3. Klik tombol “tambah user baru”	
	4. Menampilkan form tambah user
5. Mengisi form tambah user	
6. Menekan tombol “simpan”	
	7. Menyimpan data
	8. Menampilkan halaman admin kembali dan menampilkan data di tabel manajemen user.

Edit user

Name : Edit
IDE : UC003
Description : Mengubah data user
Actor : Admin

Main Course	
Aksi Aktor	**Reaksi Sistem**
1. Klik tombol “edit”	
	2. Menampilkan form edit dengan data yang sebelumnya
3. Mengisi form edit	
4. Menekan tombol “simpan”	
	5. Menyimpan data user baru
	6. Menampilkan halaman admin kembali dan menampilkan data di tabel manajemen user.

Hapus user

Name : Hapus
IDE : UC004
Description : Menghapus user
Actor : Admin

Main Course	
Aksi Aktor	**Reaksi Sistem**
1. Klik tombol “hapus”	
	2. Menampilkan message confirm
3. Menekan “oke”	
	4. Kembali Menampilkan halaman admin kembali dan menampilkan data di tabel manajemen user.

CheckIn

Name : Checkin
IDE : UC005
Description : Memasukan id parkir menggunakan card id
Actor : Pengemudi

Main Course	
Aksi Aktor	**Reaksi Sistem**
1. Men-tap card id ke device	
	2. Device mengirim data ke sistem aplikasi dan membuka palang
	3. Menampilkan data card id di tabel checkin
	4. Data berhasil disimpan

CheckOut

Name : Checkout
IDE : UC006
Description : Memasukkan id parkir yang sama setelah melakukan checkin dan melakukan pembayaran
Actor : Petugas dan Pengemudi

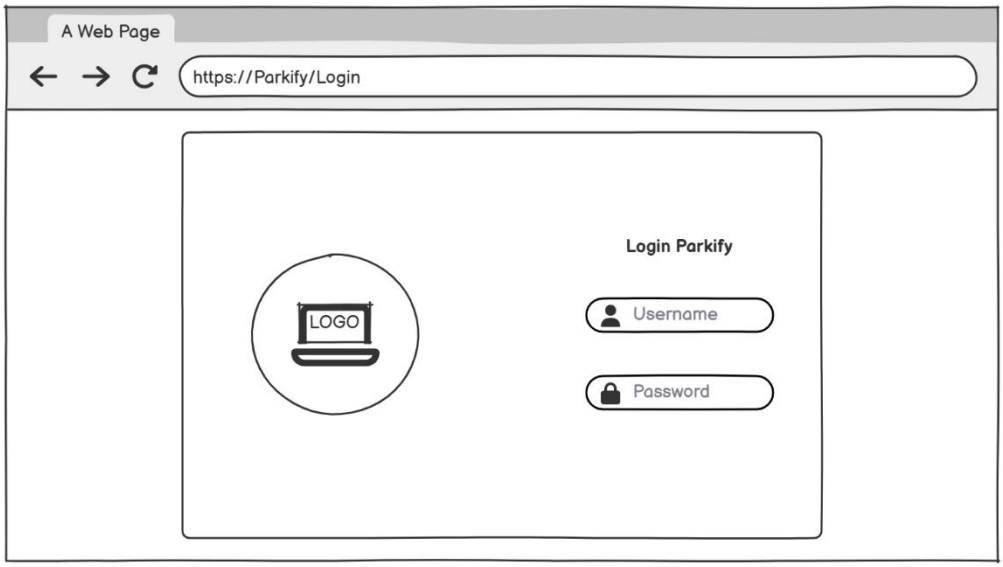
Main Course	
Aksi Aktor	**Reaksi Sistem**
1. Petugas masuk ke dashboard petugas	
	2. Menampilkan dashboard petugas
3. Pengemudi Men-tap kembali card id yang sebelumnya sudah checkin	
	4. Menampilkan card id di tabel checkout
5. Petugas melakukan pembayaran dan menekan tombol “Buka palang & cetak struk”	
	6. Device membuka palang dan aplikasi mencetak struk
	7. Menyimpan data checkout ke dalam tabel riwayat

Rekap

Name : Rekap
IDE : UC007
Description : Melihat rekap penghasilan dan laporan data parkir
Actor : Owner

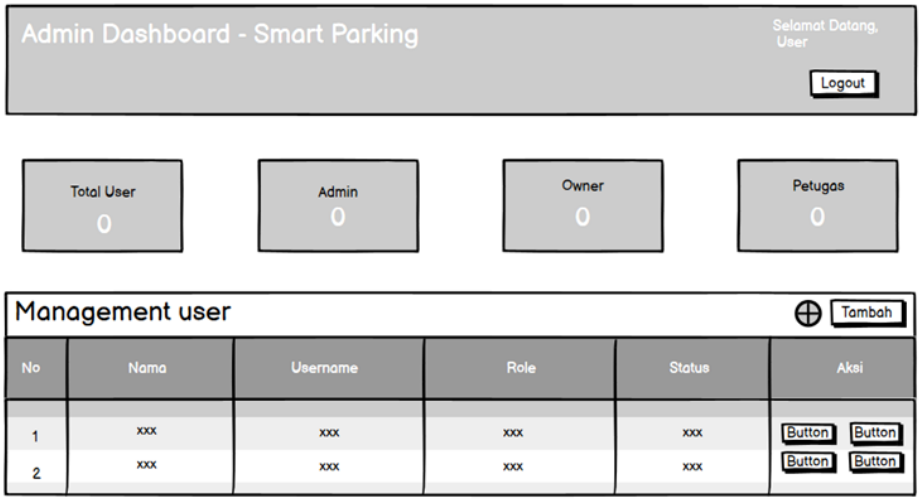
Main Course	
Aksi Aktor	**Reaksi Sistem**
1. Masuk ke halaman owner	
	2. Menampilkan halaman owner
3. Memfilter rekap sesuai tanggal yang diperlukan	
4. Menekan tombol “Tampilkan rekap”	
	5. Menampilkan data parkir yang telah selesai sesuai tanggal yang diminta.

3. Mockup



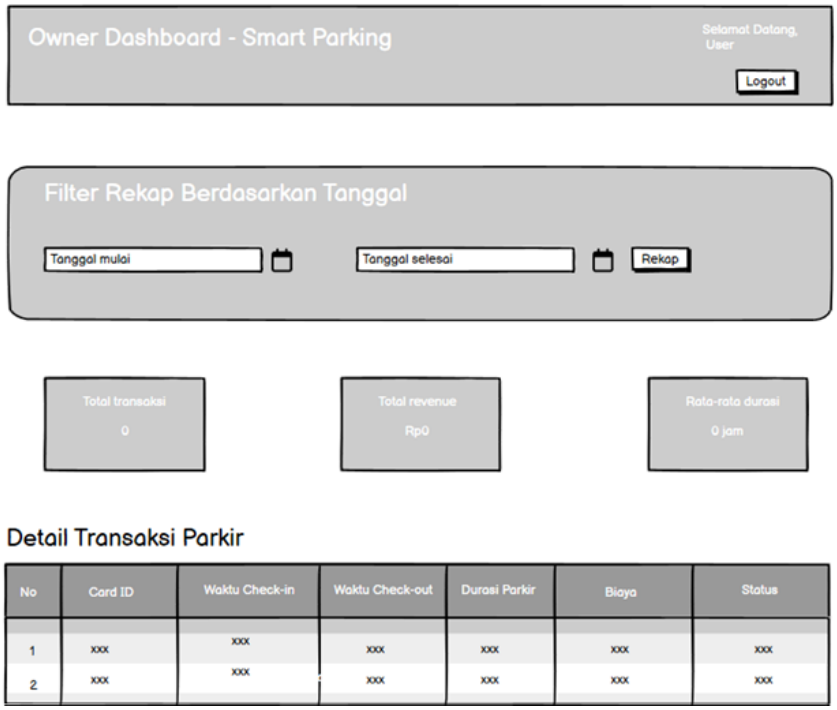
Gambar 1.18 Tampilan Login

Tampilan Sebagai Admin



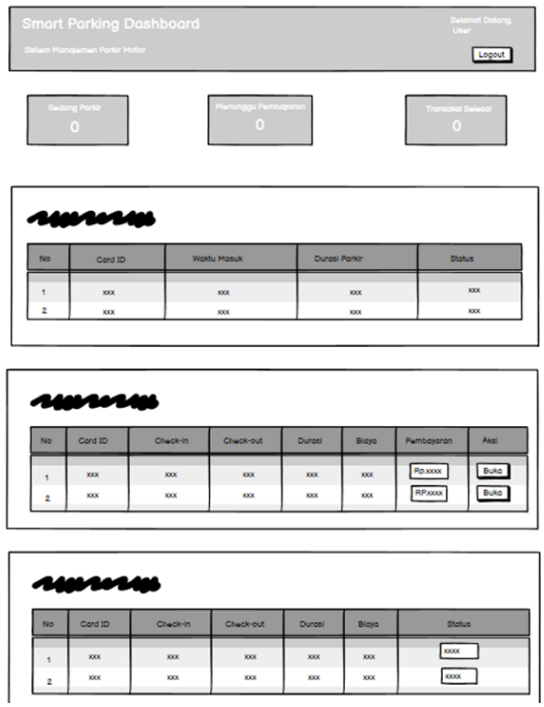
Gambar 1.19 Tampilan Dashboard Admin

Tampilan Sebagai Owner

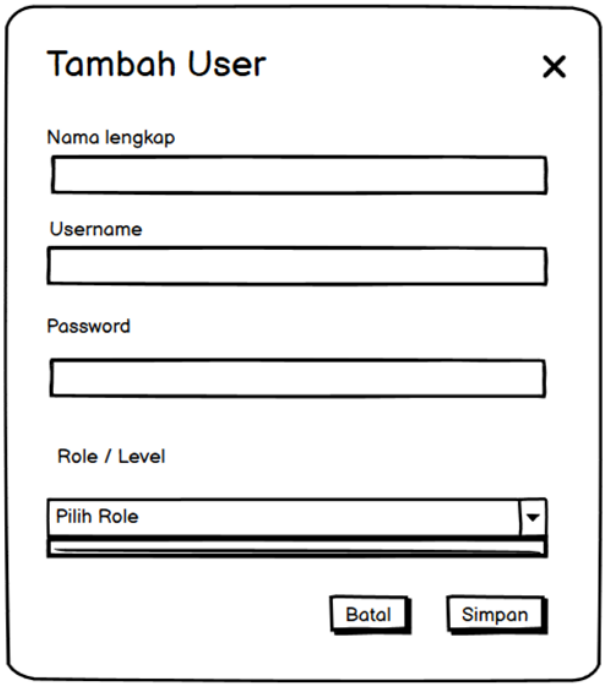


Gambar 1.20 Tampilan Dashboard Owner

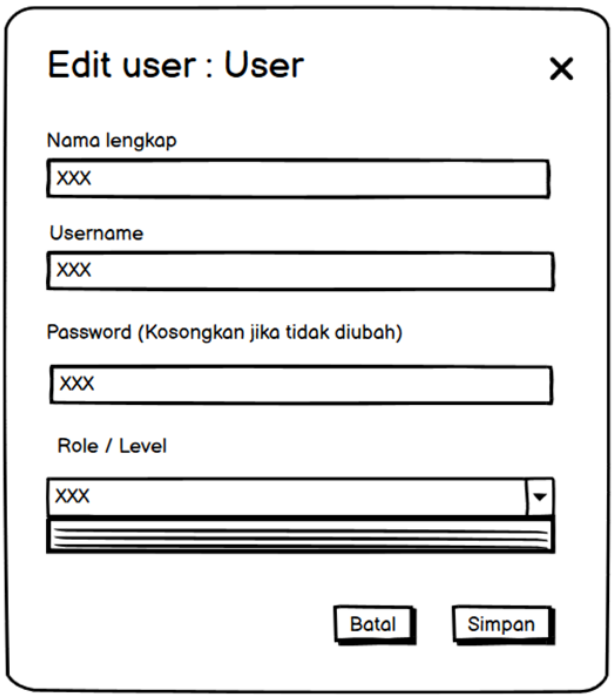
Tampilan sebagai Petugas



Gambar 1.21 Tampilan Dashboard Petugas

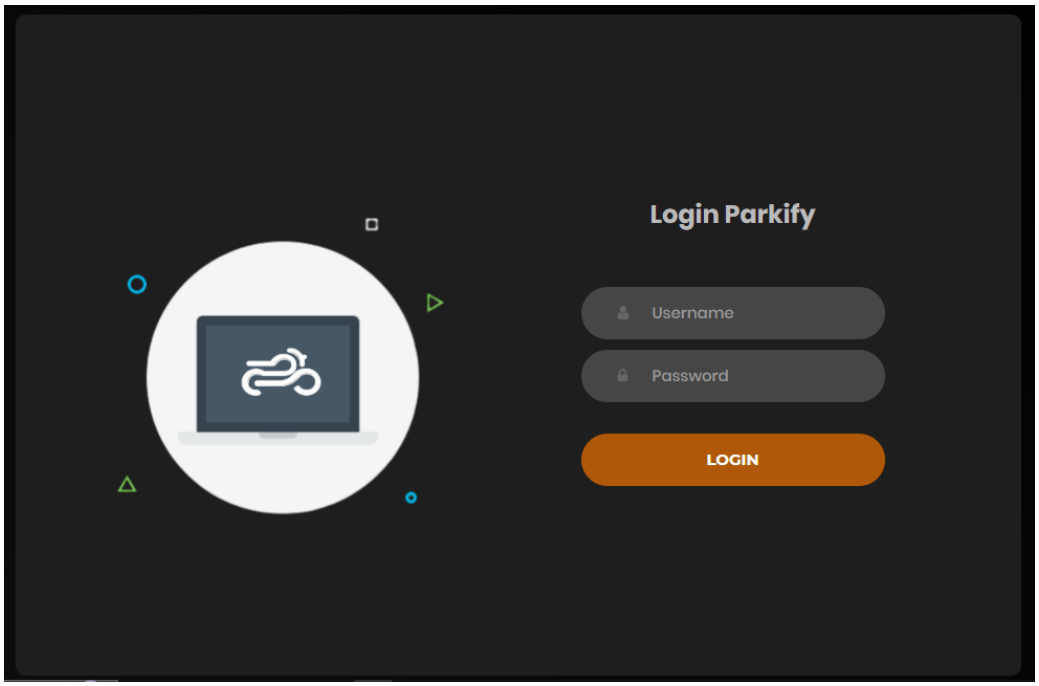


Gambar 1.23 Tampilan Tambah User

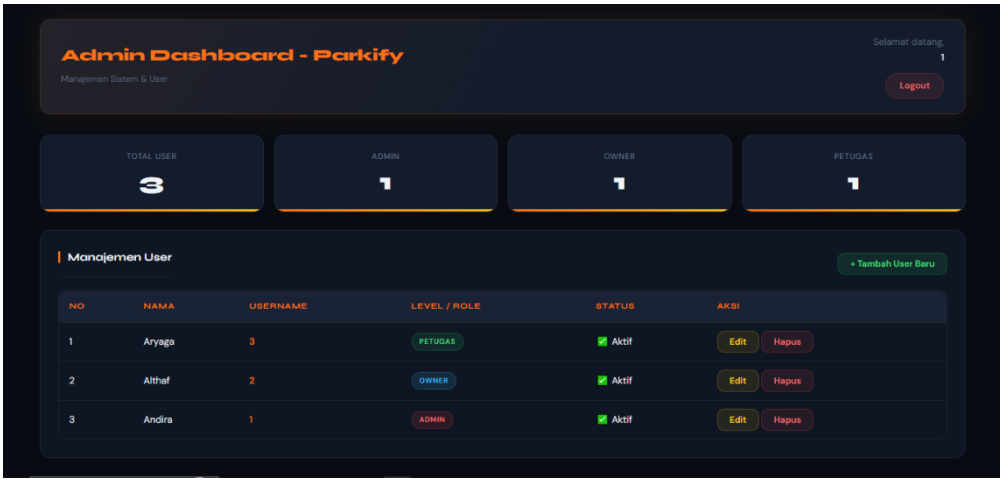


Gambar 1.24 Tampilan Ubah User

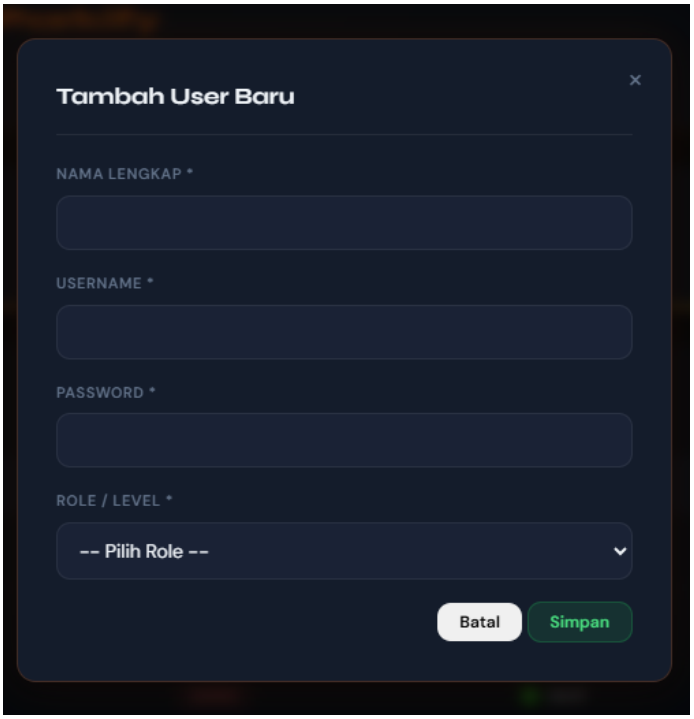
5. Implementasi UI



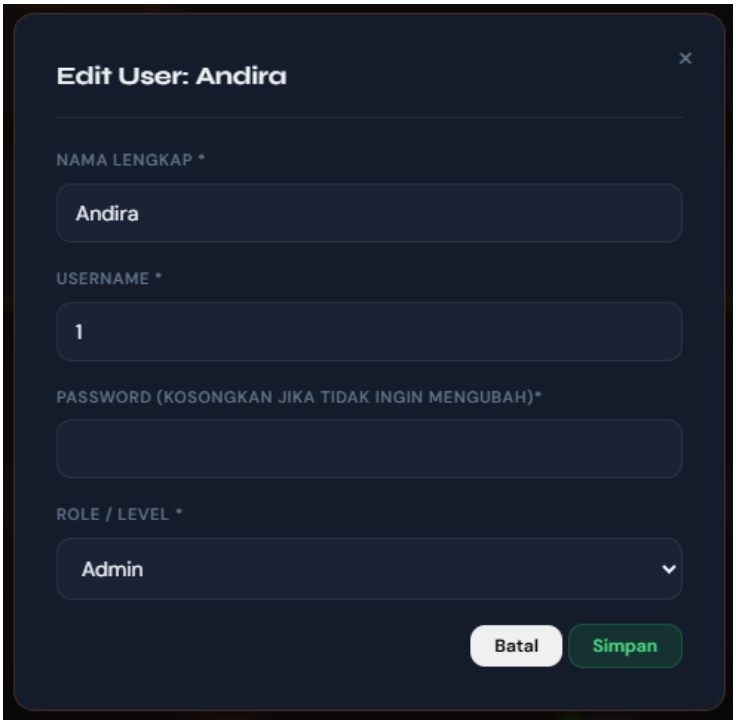
Gambar 1.25 Tampilan UI Login



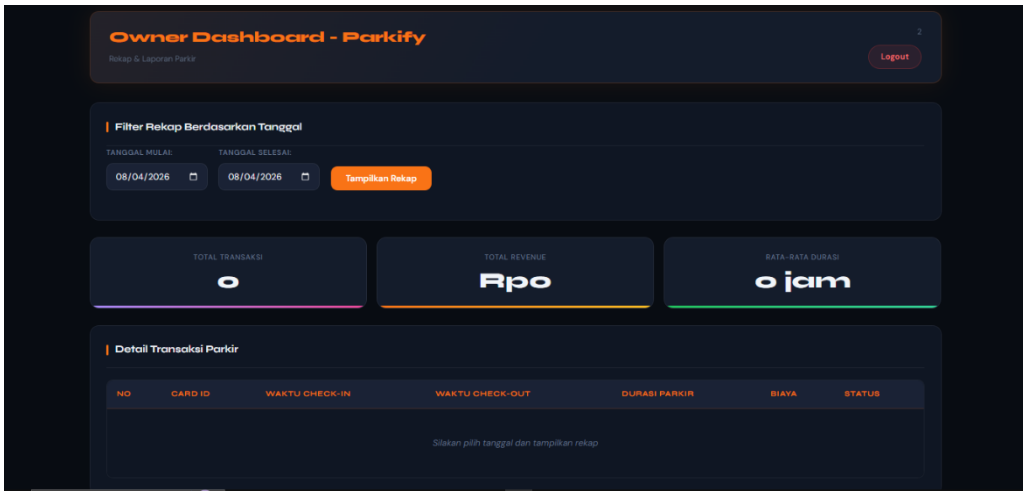
Gambar 1.26 Tampilan UI Dashboard Admin



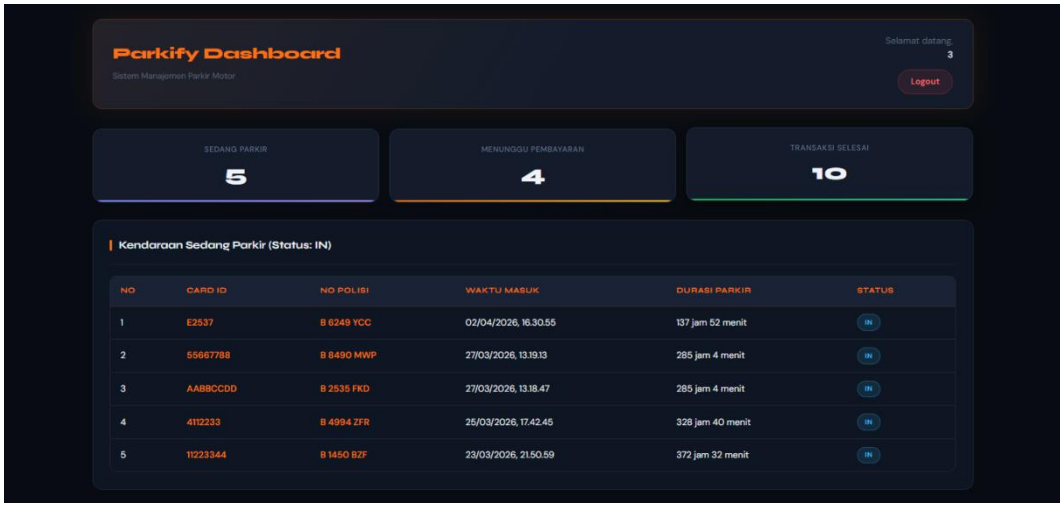
Gambar 1.27 Tampilan UI Tambah User



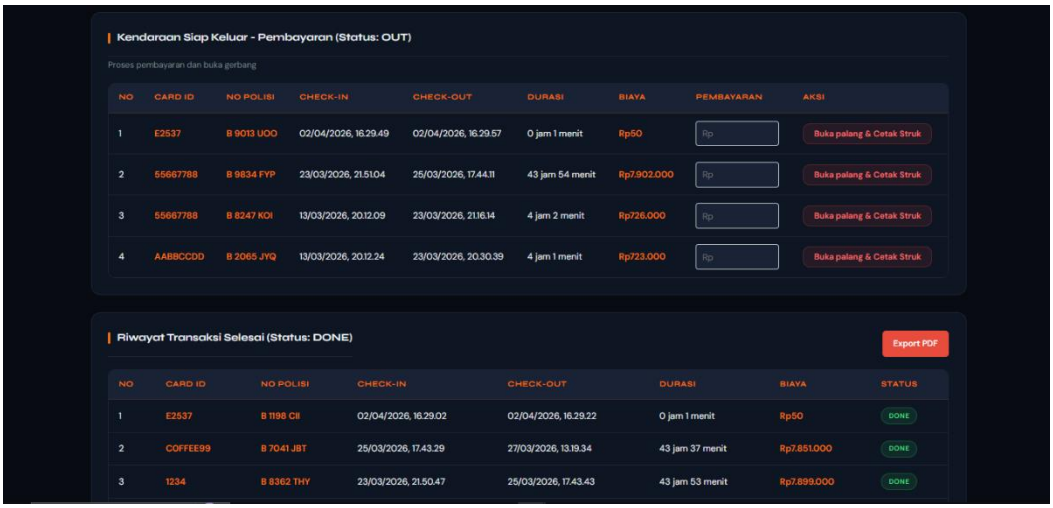
Gambar 1.28 Tampilan UI Ubah User



Gambar 1.29 Tampilan UI Dashboard Owner



Gambar 1.30 Tampilan UI Dashboard Petugas 1



Gambar 1.31 Tampilan UI Dashboard Petugas 2

10. PEMBUATAN STRUKTUR WEB (HTML)

1. halaman_petugas.php

```
<?php

if (!isset($_SESSION['username']) || $_SESSION['level'] != 'pegawai') {
    header("Location: index.php?page=login&pesan=gagal");
exit;
}

require 'config/koneksi.php';

?>
<!DOCTYPE html>
<html>
<head>
    <title>Dashboard Petugas - Smart Parking</title>
    <link rel="stylesheet" href="assets/css/style.css?v=3">
```

```
</head>
<body>
<div class="container">
    <div class="header">
        <div>
            <h1>Parkify Dashboard</h1>
            <p style="margin: 5px 0; opacity: 0.9;">Sistem Manajemen
Parkir Motor</p>
        </div>
        <div class="user-info">
            <p style="margin: 0 0 10px 0;">Selamat
datang,<br><strong><?php echo $_SESSION['username']; ?></strong></p>
            <a href="index.php?page=logout" class="logout-btn"
onclick="return confirmLogout()">Logout</a>
        </div>
    </div>

    <!-- Alert Messages -->
    <div id="alertBox" class="alert"></div>

    <!-- Statistics Cards -->
    <div class="stats" id="statsContainer">
        <div class="stat-card active">
            <div class="stat-label">Sedang Parkir</div>
            <div class="stat-number" id="countActive">0</div>
        </div>
        <div class="stat-card pending">
            <div class="stat-label">Menunggu Pembayaran</div>
            <div class="stat-number" id="countPending">0</div>
        </div>
        <div class="stat-card done">
            <div class="stat-label">Transaksi Selesai</div>
            <div class="stat-number" id="countDone">0</div>
        </div>
    </div>



    <!-- ACTIVE PARKING (IN) SECTION -->
    <div class="section">
        <h2>Kendaraan Sedang Parkir (Status: IN)</h2>
        <div class="table-wrapper">
            <table>
                <thead>
                    <tr>
                        <th>No</th>
                        <th>Card ID</th>
                        <th>No Polisi</th>
                        <th>Waktu Masuk</th>
                        <th>Durasi Parkir</th>
                        <th>Status</th>
                    </tr>
                </thead>
                <tbody id="activeTableBody">
                    <tr><td colspan="6" class="empty-message">Belum ada
kendaraan yang parkir</td></tr>
```

```
                </tbody>
            </table>
        </div>
    </div>

    <!-- PENDING CHECKOUT (OUT) SECTION -->
    <div class="section">
        <h2>Kendaraan Siap Keluar - Pembayaran (Status: OUT)</h2>
        <p style="color: #7f8c8d; margin-top: 0;">Proses pembayaran dan
buka gerbang</p>
        <div id="pendingTableWrapper" class="table-wrapper">
            <table>
                <thead>
                    <tr>
                        <th>No</th>
                        <th>Card ID</th>
                        <th>No Polisi</th>
                        <th>Check-In</th>
                        <th>Check-Out</th>
                        <th>Durasi</th>
                        <th>Biaya</th>
                        <th>Pembayaran</th>
                        <th>Aksi</th>
                    </tr>
                </thead>
                <tbody id="pendingTableBody">
                    <tr><td colspan="9" class="empty-message">Tidak ada
kendaraan yang siap keluar</td></tr>
                </tbody>
            </table>
        </div>
        <div id="paymentAlert" class="alert"></div>
    </div>

    <!-- HISTORY SECTION -->
    <div class="section">
        <div style="display:flex; justify-content:space-between; align-
items:center;">
    <h2>Riwayat Transaksi Selesai (Status: DONE)</h2>

    <a href="controller/export_pdf.php" target="_blank"
       style="background:#e74c3c; color:white; padding:8px 15px;
border-radius:5px; text-decoration:none;">
       Export PDF
    </a>
</div>
        <div class="table-wrapper">
            <table>
                <thead>
                    <tr>
                        <th>No</th>
                        <th>Card ID</th>
                        <th>No Polisi</th>
                        <th>Check-In</th>
                        <th>Check-Out</th>
                        <th>Durasi</th>
                        <th>Biaya</th>
                        <th>Status</th>
                    </tr>
```

```
                </thead>
                <tbody id="historyTableBody">
                    <tr><td colspan="8" class="empty-message">Belum ada
transaksi selesai</td></tr>
                </tbody>
            </table>
        </div>
    </div>
</div>

<script>
// Auto-focus ke input check-in
document.addEventListener('DOMContentLoaded', function() {
    const checkinInput = document.getElementById('cardIdCheckin');

    if (checkinInput) {
        checkinInput.focus();
    }

    loadAllData();
    setInterval(loadAllData, 3000);
});

// Func: Tampilkan alert
function showAlert(elementId, message, type = 'info') {
    const alertEl = document.getElementById(elementId);
    alertEl.textContent = message;
    alertEl.className = 'alert ' + type;
    alertEl.style.display = 'block';
}

// Func: Clear alert
function clearAlert(elementId) {
    const alertEl = document.getElementById(elementId);
    alertEl.className = 'alert';
    alertEl.style.display = 'none';
}

// Func: Handle Check-In
function handleCheckin() {
    const cardId =
document.getElementById('cardIdCheckin').value.trim();
    const nopol = document.getElementById('nopolCheckin').value.trim();

    if (!cardId) {
        showAlert('checkinAlert', '⚠ Silakan tap/input Card ID
terlebih dahulu', 'error');
        return;
    }

    fetch('controller/checkin.php', {
        method: 'POST',
        headers: { 'Content-Type': 'application/x-www-form-urlencoded'
},
        body: 'card_id=' + encodeURIComponent(cardId) +
      '&nopol=' + encodeURIComponent(nopol)
    })
    .then(response => response.json())
    .then(data => {
```

```
        if (data.status === 'success') {
            showAlert('checkinAlert', '✅ ' + data.message + ' (Card: '
+ data.card_id + ')', 'success');
            document.getElementById('cardIdCheckin').value = '';
            document.getElementById('cardIdCheckin').focus();
            loadAllData();
        } else {
            showAlert('checkinAlert', '❌ ' + data.message, 'error');
        }
    })
    .catch(error => {
        showAlert('checkinAlert', '❌ Error: ' + error, 'error');
    });
}
//func: logout
function confirmLogout() {
    return confirm("Apakah Anda yakin ingin logout?");
}
// Func: Handle Check-Out
function handleCheckout() {
    const cardId =
document.getElementById('cardIdCheckout').value.trim();

    if (!cardId) {
        showAlert('checkoutAlert', '⚠️ Silakan tap/input Card ID
terlebih dahulu', 'error');
        return;
    }

    fetch('controller/checkout.php', {
        method: 'POST',
        headers: { 'Content-Type': 'application/x-www-form-urlencoded'
},
        body: 'card_id=' + encodeURIComponent(cardId)
    })
    .then(response => response.json())
    .then(data => {
        if (data.status === 'success') {
            showAlert('checkoutAlert', '✅ ' + data.message + '\n💰
Biaya: ' + data.fee_display, 'success');
            document.getElementById('cardIdCheckout').value = '';
            document.getElementById('cardIdCheckout').focus();
            loadAllData();
        } else {
            showAlert('checkoutAlert', '❌ ' + data.message, 'error');
        }
    })
    .catch(error => {
        showAlert('checkoutAlert', '❌ Error: ' + error, 'error');
    });
}

// Func: Handle Payment & Gate Open
function handlePayment(transaksiId) {
    const amountInput = document.getElementById('amount_' +
transaksiId);
    const amount = amountInput.value.trim();
```

```
    if (!amount || isNaN(amount)) {
        showAlert('paymentAlert', '⚠️ Silakan input jumlah pembayaran
yang valid', 'error');
        return;
    }

    fetch('controller/handle_payment.php', {
        method: 'POST',
        headers: { 'Content-Type': 'application/x-www-form-urlencoded'
},
        body: 'transaksi_id=' + transaksiId + '&amount_paid=' + amount
    })
    .then(response => response.json())
    .then(data => {
    if (data.status === 'success') {
        showAlert('paymentAlert',
            '✅ ' + data.message + '\n💰 Kembalian: ' +
data.kembalian_display,
            'success'
        );

        window.open('controller/struk.php?id=' + transaksiId,
'_blank');

        loadAllData();
    } else {
        showAlert('paymentAlert',
            '❌ ' + data.message + '\n(Diperlukan: ' +
            (data.required_fee ? 'Rp' +
data.required_fee.toString().replace(/\B(?=(\d{3})+(?!\d))/g, '.') : '-
') + ')',
            'error'
        );
    }
})


}


// Func: Load all data (active, pending, history)
function loadAllData() {
    clearAlert('paymentAlert');

    // Load Active (IN)
    fetch('controller/get_transaksi.php?type=active')
        .then(response => response.json())
        .then(data => {
            if (data.status === 'success' && data.data.length > 0) {
                let html = '';
                data.data.forEach((row, idx) => {
                    const duration = calculateDuration(new
Date(row.checkin_time));
                    html += `
                        <tr>
                            <td>${idx + 1}</td>
                            <td><strong>${row.card_id}</strong></td>
                            <td><strong>${row.nopol ?? '-
```

```
'}</strong></td>

<td>${formatDateTime(row.checkin_time)}</td>
                            <td>${duration}</td>
                            <td><span class="status-badge status-
in">${row.status}</span></td>
                        </tr>
                    `;
                });
                document.getElementById('activeTableBody').innerHTML =
html;
                document.getElementById('countActive').textContent =
data.data.length;
            } else {
                document.getElementById('activeTableBody').innerHTML =
'<tr><td colspan="6" class="empty-message">Belum ada kendaraan yang
parkir</td></tr>';
                document.getElementById('countActive').textContent =
'0';
            }
        });

    // Load Pending (OUT) - SKIP jika ada input biaya yang sedang di-
focus
    const focusedElement = document.activeElement;
    const isFocusedOnPaymentInput = focusedElement && focusedElement.id
&& focusedElement.id.startsWith('amount_');

    if (!isFocusedOnPaymentInput) {
        fetch('controller/get_transaksi.php?type=pending')
            .then(response => response.json())
            .then(data => {
                if (data.status === 'success' && data.data.length > 0)
{
                    let html = '';
                    data.data.forEach((row, idx) => {
                        const duration = (row.duration /
60).toFixed(2);
                        html += `
                            <tr>
                                <td>${idx + 1}</td>

<td><strong>${row.card_id}</strong></td>
                                <td><strong>${row.nopol ?? '-
'}</strong></td>

<td>${formatDateTime(row.checkin_time)}</td>

<td>${formatDateTime(row.checkout_time)}</td>
                                <td>${parseInt(row.duration / 60)} jam
${row.duration % 60} menit</td>

<td><strong>${row.fee_display}</strong></td>
                                <td>
                                    <input type="number"
id="amount_${row.id}" placeholder="Rp" min="0" style="width: 120px
!important; padding: 8px !important; border: 1px solid #bdc3c7
!important; border-radius: 4px !important; flex: none !important; min-
width: auto !important;">
```

```
                                </td>
                                <td>
                                    <button
onclick="handlePayment(${row.id})" class="btn btn-danger" style="width:
auto; padding: 8px 15px;">Buka palang & Cetak Struk</button>
                                </td>
                            </tr>
                        `;
                    });

document.getElementById('pendingTableBody').innerHTML = html;
                    document.getElementById('countPending').textContent
= data.data.length;
                } else {

document.getElementById('pendingTableBody').innerHTML = '<tr><td
colspan="9" class="empty-message">Tidak ada kendaraan yang siap
keluar</td></tr>';
                    document.getElementById('countPending').textContent
= '0';
                }
            });
    }

    // Load History (DONE)
    fetch('controller/get_transaksi.php?type=history')
        .then(response => response.json())
        .then(data => {
            if (data.status === 'success' && data.data.length > 0) {
                let html = '';
                data.data.forEach((row, idx) => {
                    html += `
                        <tr>
                            <td>${idx + 1}</td>
                            <td><strong>${row.card_id}</strong></td>
                            <td><strong>${row.nopol ?? '-
'}</strong></td>

<td>${formatDateTime(row.checkin_time)}</td>

<td>${formatDateTime(row.checkout_time)}</td>
                            <td>${row.duration_display}</td>

<td><strong>${row.fee_display}</strong></td>
                            <td><span class="status-badge status-
done">${row.status}</span></td>
                        </tr>
                    `;
                });
                document.getElementById('historyTableBody').innerHTML =
html;
                document.getElementById('countDone').textContent =
data.data.length;
            } else {
                document.getElementById('historyTableBody').innerHTML =
'<tr><td colspan="8" class="empty-message">Belum ada transaksi
selesai</td></tr>';
                document.getElementById('countDone').textContent = '0';
            }
```

```
        });
}

// Func: Format datetime
function formatDateTime(dateString) {
    const date = new Date(dateString);
    const options = { year: 'numeric', month: '2-digit', day: '2-
digit', hour: '2-digit', minute: '2-digit', second: '2-digit' };
    return date.toLocaleDateString('id-ID', options);
}

// Func: Calculate duration from check-in time until now
function calculateDuration(checkinTime) {
    const now = new Date();
    const diff = now - checkinTime;
    const hours = Math.floor(diff / (1000 * 60 * 60));
    const minutes = Math.floor((diff / (1000 * 60)) % 60);
    return `${hours} jam ${minutes} menit`;
}

// Allow Enter key to submit
const checkinInput = document.getElementById('cardIdCheckin');
if (checkinInput) {
    checkinInput.addEventListener('keypress', function(e) {
        if (e.key === 'Enter') handleCheckin();
    });
}

const checkoutInput = document.getElementById('cardIdCheckout');
if (checkoutInput) {
    checkoutInput.addEventListener('keypress', function(e) {
        if (e.key === 'Enter') handleCheckout();
    });
}
</script>
</body>
</html>
```

11. PEMBUATAN STRUCTURE DEVICES (ino)

1. Parkify.ino

```
#include <Arduino.h>
#include <WiFi.h>
#include <PubSubClient.h>
#include <SPI.h>
#include <MFRC522.h>
#include <ESP32Servo.h>
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

/* ================= WIFI & MQTT ================= */
const char *ssid = "Realme C25";
const char *password = "uwaaahhh";

const char *mqtt_server = "192.168.190.99"; // ← IP komputer (hasil
ipconfig)
const int   mqtt_port   = 1883;
/* ================= MQTT TOPIC ================= */
#define TOPIC_ENTRY_RFID "parking/dira/entry/rfid"
#define TOPIC_EXIT_RFID "parking/dira/exit/rfid"
#define TOPIC_ENTRY_SERVO "parking/dira/entry/servo"
#define TOPIC_EXIT_SERVO "parking/dira/exit/servo"
#define TOPIC_LCD "parking/dira/lcd"

/* ================= RFID ================= */
#define RST_ENTRY_PIN 27
#define SS_ENTRY_PIN 5
#define RST_EXIT_PIN 17
#define SS_EXIT_PIN 15

MFRC522 rfidEntry(SS_ENTRY_PIN, RST_ENTRY_PIN);
MFRC522 rfidExit(SS_EXIT_PIN, RST_EXIT_PIN);

/* ================= SERVO ================= */
#define SERVO_ENTRY_PIN 32
#define SERVO_EXIT_PIN 33

Servo servoEntry;
Servo servoExit;

/* ================= OLED ================= */
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
#define OLED_RESET -1

Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire,
OLED_RESET);

/* ================= MQTT ================= */
WiFiClient espClient;
PubSubClient client(espClient);
```

```
void connectWiFi()
{
      WiFi.begin(ssid, password);
      while (WiFi.status() != WL_CONNECTED)
      {
            delay(500);
      }
}

String getMqttClientId()
{
      uint64_t chipid = ESP.getEfuseMac();
      char id[50];
      snprintf(id, sizeof(id), "smartparking-%04X%08X",
                  (uint16_t)(chipid >> 32),
                  (uint32_t)chipid);
      return String(id);
}

void reconnectMQTT()
{
      while (!client.connected())
      {
            if (client.connect(getMqttClientId().c_str()))
            {
                  client.subscribe(TOPIC_ENTRY_SERVO);
                  client.subscribe(TOPIC_EXIT_SERVO);
                  client.subscribe(TOPIC_LCD);
            }
            else
            {
                  delay(2000);
            }
      }
}

/* ================= SERVO CONTROL ================= */
void openServo(Servo &servo)
{
      servo.write(90);
      delay(1000);
      servo.write(0);
}

/* ================= OLED DISPLAY ================= */
void showOLED(String line1, String line2)
{
      display.clearDisplay();
      display.setTextSize(1);
      display.setTextColor(SSD1306_WHITE);

      display.setCursor(0, 10);
      display.println(line1);

      display.setTextSize(1);
      display.setCursor(0, 40);
```

```
      display.println(line2);

      display.display();
}

/* ================= MQTT CALLBACK ================= */
void callback(char *topic, byte *payload, unsigned int length)
{
  String message = "";
  for (int i = 0; i < length; i++) {
    message += (char)payload[i];
  }

  Serial.println("Pesan Masuk [" + String(topic) + "]: " + message);

  // Logika Palang Masuk
  if (String(topic) == TOPIC_ENTRY_SERVO) {
    if (message == "OPEN") {
      showOLED("ACCESS GRANTED", "Selamat Datang!"); // Tampilkan
Ucapan
      servoEntry.write(90);
      delay(3000);
      servoEntry.write(0);
      showOLED("Parkify", "System Ready");
    }
  }

  // Logika Palang Keluar
      if (String(topic) == TOPIC_EXIT_SERVO) {
    if (message == "OPEN") {
      showOLED("LUNAS", "Silahkan Jalan");
      servoExit.write(90);
      delay(3000);
      servoExit.write(0);
      showOLED("Parkify", "System Ready");
    }
}
 // if (String(topic) == TOPIC_EXIT_SERVO) {
   // if (message == "OPEN") {
     // showOLED("ACCESS GRANTED", "Selamat Jalan!"); // Tampilkan
Ucapan
      //servoExit.write(90);
      //delay(3000);
      //servoExit.write(0);
      //showOLED("Smart Parking", "System Ready");
   // }
 // }

  // Menampilkan Durasi & Biaya dari Bridge
  if (String(topic) == TOPIC_LCD) {
    // Bridge akan mengirim format: "Durasi: 2 Jam|Total: Rp4.000"
    int separator = message.indexOf('|');
    if (separator != -1) {
      String line1 = message.substring(0, separator);
      String line2 = message.substring(separator + 1);
      showOLED(line1, line2);
```

```
    } else {
      showOLED("INFO", message);
    }
  }
}

/* ================= RFID READ ================= */
String readRFID(MFRC522 &rfid)
{
      if (!rfid.PICC_IsNewCardPresent() || !rfid.PICC_ReadCardSerial())
      {
            return "";
      }

      String uid = "";
      for (byte i = 0; i < rfid.uid.size; i++)
      {
            uid += String(rfid.uid.uidByte[i], HEX);
      }
      uid.toUpperCase();

      rfid.PICC_HaltA();
      rfid.PCD_StopCrypto1();

      return uid;
}

/* ================= SETUP ================= */
void setup()
{
      Serial.begin(115200);

      // RFID Setup
      SPI.begin();
      rfidEntry.PCD_Init();
      rfidExit.PCD_Init();

      // Servo Setup
      servoEntry.attach(SERVO_ENTRY_PIN);
      servoExit.attach(SERVO_EXIT_PIN);
      servoEntry.write(0);
      servoExit.write(0);

      // OLED Init
      Wire.begin(21, 22);
      if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C))
      {
            Serial.println("OLED not found");
            while (true)
                  ;
      }

      showOLED("Parkify", "System Ready");

      // WiFi & MQTT Setup
      connectWiFi();
```

```
	client.setServer(mqtt_server, mqtt_port);
	client.setCallback(callback);
}

/* ================= LOOP ================= */
void loop()
{
	if (!client.connected())
	{
		reconnectMQTT();
	}
	client.loop();

	// ENTRY
	String uidEntry = readRFID(rfidEntry);
	if (uidEntry != "")
	{
		String payload = "{\"rfid\":\"" + uidEntry + "\"}";
		client.publish(TOPIC_ENTRY_RFID, payload.c_str());
		Serial.println("ENTRY : " + payload);
		delay(2000);
	}

	// EXIT
	String uidExit = readRFID(rfidExit);
	if (uidExit != "")
	{
		String payload = "{\"rfid\":\"" + uidExit + "\"}";
		client.publish(TOPIC_EXIT_RFID, payload.c_str());
		Serial.println("EXIT : " + payload);
		delay(2000);
	}
}
```

12. PEMBUATAN STYLE WEB (CSS)

1. style.css

```
/* ============================================================
   PARKIFY - Dark Theme (Black & Orange)
   Primary: #0d0d0d / #111827  |  Accent: #f97316 / #ea580c
   ============================================================ */

@import
url('https://fonts.googleapis.com/css2?family=Syne:wght@400;600;700;800&family
=DM+Sans:ital,opsz,wght@0,9..40,300;0,9..40,400;0,9..40,500;0,9..40,600;1,9..4
0,400&display=swap');

:root {
  --bg-base:      #080c12;
  --bg-surface:   #0f1623;
  --bg-card:      #141c2b;
  --bg-elevated:  #1a2235;
  --accent:       #f97316;
  --accent-dim:   #ea580c;
  --accent-glow:  rgba(249,115,22,0.18);
  --accent-faint: rgba(249,115,22,0.07);
  --text-primary: #f1f5f9;
  --text-muted:   #64748b;
  --text-faint:   #334155;
  --border:       rgba(249,115,22,0.2);
  --border-faint: rgba(255,255,255,0.07);
  --danger:       #ef4444;
  --success:      #22c55e;
  --warning:      #eab308;
  --info:         #38bdf8;
  --radius:       10px;
  --radius-lg:    14px;
  --shadow:       0 4px 24px rgba(0,0,0,0.55);
  --shadow-glow:  0 0 32px rgba(249,115,22,0.1);
}

/* ─── Reset ──────────────────────────────────────────────── */
*, *::before, *::after { margin: 0; padding: 0; box-sizing: border-box; }

body {
  font-family: 'DM Sans', sans-serif;
  background: var(--bg-base);
  color: var(--text-primary);
  min-height: 100vh;
  font-size: 14px;
  line-height: 1.6;
}

::-webkit-scrollbar { width: 6px; height: 6px; }
::-webkit-scrollbar-track { background: var(--bg-base); }
::-webkit-scrollbar-thumb { background: #1e2a3a; border-radius: 99px; }
::-webkit-scrollbar-thumb:hover { background: var(--accent-dim); }

/* ─── Container ──────────────────────────────────────────── */
.container {
  max-width: 1300px;
  margin: 0 auto;
```

```
  padding: 24px 20px;
}

/* ─── Header ─────────────────────────────────────────────────── */
.header {
  background: var(--bg-card);
  border: 1px solid var(--border);
  padding: 20px 28px;
  border-radius: var(--radius-lg);
  margin-bottom: 28px;
  display: flex;
  justify-content: space-between;
  align-items: center;
  box-shadow: var(--shadow-glow);
  position: relative;
  overflow: hidden;
}

.header::before {
  content: '';
  position: absolute;
  inset: 0;
  background: linear-gradient(120deg, rgba(249,115,22,0.07) 0%, transparent
55%);
  pointer-events: none;
}

.header h1 {
  font-family: 'Syne', sans-serif;
  font-size: 22px;
  font-weight: 800;
  color: var(--accent);
  letter-spacing: -0.3px;
  margin: 0;
  text-align: left;
}

.header p {
  color: var(--text-muted);
  font-size: 12px;
  margin: 5px 0 0;
  opacity: 0.9;
}

.user-info { text-align: right; }

.user-info p {
  color: var(--text-muted);
  font-size: 13px;
  margin: 0 0 10px;
}

.user-info strong { color: var(--text-primary); }

/* ─── Logout Button ──────────────────────────────────────────── */
.logout-btn {
  display: inline-block;
  background: rgba(239,68,68,0.1);
  color: #f87171;
```

```
  border: 1px solid rgba(239,68,68,0.25);
  padding: 7px 18px;
  border-radius: 99px;
  text-decoration: none;
  font-size: 12px;
  font-weight: 600;
  cursor: pointer;
  transition: all 0.2s;
  letter-spacing: 0.3px;
}

.logout-btn:hover {
  background: #ef4444;
  color: #fff;
  border-color: #ef4444;
  box-shadow: 0 0 14px rgba(239,68,68,0.35);
}

/* ─── Section ──────────────────────────────────────────────── */
.section {
  background: var(--bg-surface);
  border: 1px solid var(--border-faint);
  border-radius: var(--radius-lg);
  padding: 24px;
  margin-bottom: 24px;
}

.section h2 {
  font-family: 'Syne', sans-serif;
  font-size: 15px;
  font-weight: 700;
  color: var(--text-primary);
  margin: 0 0 2px;
  padding-bottom: 14px;
  border-bottom: 1px solid var(--border-faint);
  display: flex;
  align-items: center;
  gap: 10px;
}

.section h2::before {
  content: '';
  display: inline-block;
  width: 3px;
  height: 16px;
  background: var(--accent);
  border-radius: 4px;
  flex-shrink: 0;
}

.section > p {
  color: var(--text-muted);
  font-size: 12px;
  margin: 8px 0 16px;
}

/* ─── Stats Grid ───────────────────────────────────────────── */
.stats {
  display: grid;
```

```
  grid-template-columns: repeat(auto-fit, minmax(150px, 1fr));
  gap: 14px;
  margin-bottom: 24px;
}

.stat-card {
  background: var(--bg-card);
  border: 1px solid var(--border-faint);
  border-radius: var(--radius-lg);
  padding: 20px 16px;
  text-align: center;
  position: relative;
  overflow: hidden;
  transition: transform 0.2s, box-shadow 0.2s;
  /* override inline gradient styles */
  background: var(--bg-card) !important;
  color: var(--text-primary) !important;
}

.stat-card:hover {
  transform: translateY(-2px);
  box-shadow: 0 8px 28px rgba(0,0,0,0.45);
}

.stat-card::after {
  content: '';
  position: absolute;
  bottom: 0; left: 0; right: 0;
  height: 3px;
  background: linear-gradient(90deg, var(--accent), #fbbf24);
  border-radius: 0 0 4px 4px;
}

.stat-card.active::after    { background: linear-gradient(90deg, #818cf8,
#a78bfa); }
.stat-card.pending::after   { background: linear-gradient(90deg, #f97316,
#fbbf24); }
.stat-card.done::after      { background: linear-gradient(90deg, #22c55e,
#34d399); }
.stat-card.transactions::after { background: linear-gradient(90deg, #a78bfa,
#ec4899); }
.stat-card.revenue::after   { background: linear-gradient(90deg, #f97316,
#fbbf24); }
.stat-card.duration::after  { background: linear-gradient(90deg, #22c55e,
#34d399); }

.stat-number {
  font-family: 'Syne', sans-serif;
  font-size: 32px;
  font-weight: 800;
  color: var(--text-primary);
  margin: 10px 0 4px;
  line-height: 1;
}

.stat-label {
  font-size: 11px;
  color: var(--text-muted);
  font-weight: 500;
```

```
  text-transform: uppercase;
  letter-spacing: 0.6px;
}

/* ─── Tables ──────────────────────────────────────────────── */
.table-wrapper {
  overflow-x: auto;
  border-radius: var(--radius);
  border: 1px solid var(--border-faint);
  margin-top: 16px;
}

table {
  width: 100%;
  border-collapse: collapse;
  margin: 0;
}

table th {
  background: var(--bg-elevated);
  color: var(--accent);
  padding: 12px 14px;
  text-align: left;
  font-family: 'Syne', sans-serif;
  font-size: 11px;
  font-weight: 700;
  text-transform: uppercase;
  letter-spacing: 0.7px;
  white-space: nowrap;
  border-bottom: 1px solid var(--border);
}

table td {
  padding: 11px 14px;
  border-bottom: 1px solid var(--border-faint);
  color: var(--text-primary);
  font-size: 13px;
  background: var(--bg-surface);
}

table tr:last-child td { border-bottom: none; }

table tr:hover td { background: var(--accent-faint); }

table td strong { color: var(--accent); font-weight: 600; }

/* ─── Status Badges ──────────────────────────────────────── */
.status-badge {
  display: inline-block;
  padding: 3px 10px;
  border-radius: 99px;
  font-size: 10px;
  font-weight: 700;
  text-transform: uppercase;
  letter-spacing: 0.6px;
}

.status-in     { background: rgba(56,189,248,0.12);  color: #38bdf8; border:
1px solid rgba(56,189,248,0.25); }
```

```
.status-out    { background: rgba(239,68,68,0.12);   color: #f87171; border:
1px solid rgba(239,68,68,0.25); }
.status-done   { background: rgba(34,197,94,0.12);   color: #4ade80; border:
1px solid rgba(34,197,94,0.25); }
.status-admin  { background: rgba(239,68,68,0.12);   color: #f87171; border:
1px solid rgba(239,68,68,0.25); }
.status-owner  { background: rgba(56,189,248,0.12);  color: #38bdf8; border:
1px solid rgba(56,189,248,0.25); }
.status-pegawai { background: rgba(34,197,94,0.12);  color: #4ade80; border:
1px solid rgba(34,197,94,0.25); }

/* ─── Buttons ─────────────────────────────────────────────── */
.btn {
  display: inline-flex;
  align-items: center;
  justify-content: center;
  gap: 6px;
  padding: 8px 16px;
  border: none;
  border-radius: var(--radius);
  cursor: pointer;
  font-size: 12px;
  font-weight: 600;
  font-family: 'DM Sans', sans-serif;
  transition: all 0.2s;
  text-decoration: none;
  white-space: nowrap;
  line-height: 1;
}

.btn-primary   { background: var(--accent); color: #fff; }
.btn-primary:hover { background: var(--accent-dim); box-shadow: 0 0 16px var(-
-accent-glow); }

.btn-success   { background: rgba(34,197,94,0.12); color: #4ade80; border: 1px
solid rgba(34,197,94,0.25); }
.btn-success:hover { background: #22c55e; color: #fff; border-color: #22c55e;
}

.btn-danger    { background: rgba(239,68,68,0.1); color: #f87171; border: 1px
solid rgba(239,68,68,0.25); }
.btn-danger:hover { background: #ef4444; color: #fff; border-color: #ef4444; }

.btn-warning   { background: rgba(234,179,8,0.1); color: #facc15; border: 1px
solid rgba(234,179,8,0.25); }
.btn-warning:hover { background: #eab308; color: #fff; border-color: #eab308;
}

/* ─── Alerts ──────────────────────────────────────────────── */
.alert {
  padding: 12px 16px;
  border-radius: var(--radius);
  margin-bottom: 16px;
  font-size: 13px;
  font-weight: 500;
  display: none;
  white-space: pre-line;
}
```

```
.alert.success {
  background: rgba(34,197,94,0.1);
  color: #4ade80;
  border: 1px solid rgba(34,197,94,0.2);
  display: block;
}

.alert.error {
  background: rgba(239,68,68,0.1);
  color: #f87171;
  border: 1px solid rgba(239,68,68,0.2);
  display: block;
}

.alert.info {
  background: rgba(56,189,248,0.1);
  color: #7dd3fc;
  border: 1px solid rgba(56,189,248,0.2);
  display: block;
}

/* ─── Empty Message ──────────────────────────────────────── */
.empty-message {
  text-align: center;
  color: var(--text-muted);
  padding: 40px 20px;
  font-style: italic;
  font-size: 13px;
  background: var(--bg-surface) !important;
}

/* ─── Input Groups (Petugas Check-In/Out) ─────────────────── */
.input-group {
  display: flex;
  gap: 10px;
  margin-bottom: 16px;
  flex-wrap: wrap;
  align-items: center;
}

.input-group input {
  flex: 1;
  min-width: 180px;
  padding: 10px 14px;
  background: var(--bg-elevated);
  border: 1px solid var(--border-faint);
  border-radius: var(--radius);
  color: var(--text-primary);
  font-size: 14px;
  font-family: 'DM Sans', sans-serif;
  outline: none;
  transition: border-color 0.2s, box-shadow 0.2s;
}

.input-group input:focus {
  border-color: var(--accent);
  box-shadow: 0 0 0 3px var(--accent-glow);
}
```

```
.input-group input::placeholder { color: var(--text-faint); }

.input-group button {
  padding: 10px 20px;
  background: var(--accent);
  color: #fff;
  border: none;
  border-radius: var(--radius);
  cursor: pointer;
  font-weight: 600;
  font-family: 'DM Sans', sans-serif;
  font-size: 13px;
  transition: all 0.2s;
}

.input-group button:hover {
  background: var(--accent-dim);
  box-shadow: 0 0 16px var(--accent-glow);
}

.input-group button.payment {
  background: rgba(239,68,68,0.12);
  color: #f87171;
  border: 1px solid rgba(239,68,68,0.25);
}

.input-group button.payment:hover {
  background: #ef4444;
  color: #fff;
}

/* ─── Inline Payment Input in Table ──────────────────────────── */
table input[type="number"] {
  background: var(--bg-elevated) !important;
  border: 1px solid var(--border-faint) !important;
  border-radius: var(--radius) !important;
  color: var(--text-primary) !important;
  padding: 8px 10px !important;
  font-family: 'DM Sans', sans-serif !important;
  font-size: 13px !important;
  outline: none;
  transition: border-color 0.2s, box-shadow 0.2s;
  width: 120px !important;
  flex: none !important;
  min-width: auto !important;
}

table input[type="number"]:focus {
  border-color: var(--accent) !important;
  box-shadow: 0 0 0 3px var(--accent-glow) !important;
}

/* ─── Filter Group (Owner) ─────────────────────────────────── */
.filter-group {
  display: flex;
  gap: 16px;
  margin-bottom: 20px;
  flex-wrap: wrap;
  align-items: flex-end;
```

```
}

.filter-group label {
  display: flex;
  flex-direction: column;
  gap: 6px;
  font-size: 11px;
  font-weight: 600;
  color: var(--text-muted);
  text-transform: uppercase;
  letter-spacing: 0.5px;
}

.filter-group input[type="date"] {
  padding: 9px 12px;
  background: var(--bg-elevated);
  border: 1px solid var(--border-faint);
  border-radius: var(--radius);
  color: var(--text-primary);
  font-size: 13px;
  font-family: 'DM Sans', sans-serif;
  min-width: 150px;
  outline: none;
  transition: border-color 0.2s, box-shadow 0.2s;
  color-scheme: dark;
}

.filter-group input[type="date"]:focus {
  border-color: var(--accent);
  box-shadow: 0 0 0 3px var(--accent-glow);
}

.filter-group button {
  padding: 9px 22px;
  background: var(--accent);
  color: #fff;
  border: none;
  border-radius: var(--radius);
  cursor: pointer;
  font-weight: 600;
  font-family: 'DM Sans', sans-serif;
  font-size: 13px;
  transition: all 0.2s;
}

.filter-group button:hover {
  background: var(--accent-dim);
  box-shadow: 0 0 16px var(--accent-glow);
}

/* ─── Modal ──────────────────────────────────────────────────── */
.modal {
  display: none;
  position: fixed;
  z-index: 9999;
  inset: 0;
  background: rgba(0,0,0,0.72);
  backdrop-filter: blur(5px);
  justify-content: center;
```

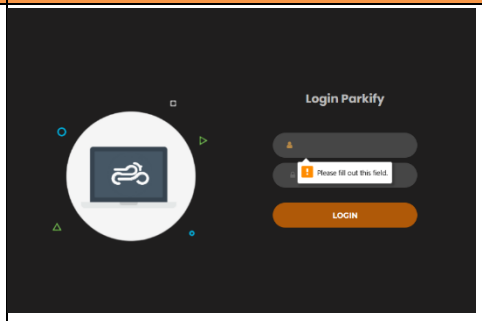
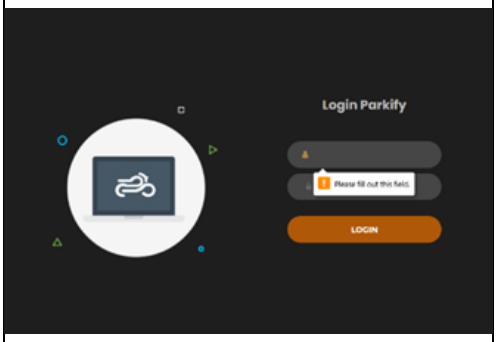
```
  align-items: center;
}

.modal.show {
  display: flex;
}

.modal-content {
  background: var(--bg-card);
  border: 1px solid var(--border);
  border-radius: var(--radius-lg);
  padding: 28px;
  width: 90%;
  max-width: 480px;
  max-height: 90vh;
  overflow-y: auto;
  box-shadow: var(--shadow), var(--shadow-glow);
  position: relative;
  animation: modalIn 0.22s ease;
}

@keyframes modalIn {
  from { opacity: 0; transform: scale(0.96) translateY(10px); }
  to   { opacity: 1; transform: scale(1) translateY(0); }
}

.modal-content h2 {
  font-family: 'Syne', sans-serif;
  font-size: 17px;
  font-weight: 700;
  color: var(--text-primary);
  margin: 0 0 20px;
  padding-bottom: 14px;
  border-bottom: 1px solid var(--border-faint);
}

.close {
  position: absolute;
  top: 18px;
  right: 20px;
  font-size: 22px;
  color: var(--text-muted);
  cursor: pointer;
  transition: color 0.2s;
  float: none;
  line-height: 1;
}
.close:hover { color: var(--accent); }

/* ─── Form Groups (inside Modal) ──────────────────────────── */
.form-group {
  margin-bottom: 16px;
}

.form-group label {
  display: block;
  font-size: 11px;
  font-weight: 600;
  color: var(--text-muted);
```

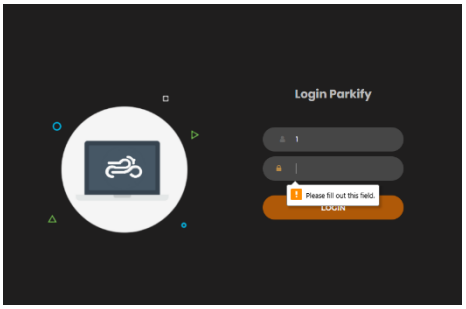
```
  text-transform: uppercase;
  letter-spacing: 0.5px;
  margin-bottom: 6px;
}

.form-group input,
.form-group select {
  width: 100%;
  padding: 10px 12px;
  background: var(--bg-elevated);
  border: 1px solid var(--border-faint);
  border-radius: var(--radius);
  color: var(--text-primary);
  font-size: 14px;
  font-family: 'DM Sans', sans-serif;
  outline: none;
  transition: border-color 0.2s, box-shadow 0.2s;
  box-sizing: border-box;
}

.form-group input:focus,
.form-group select:focus {
  border-color: var(--accent);
  box-shadow: 0 0 0 3px var(--accent-glow);
}

.form-group select option {
  background: var(--bg-elevated);
  color: var(--text-primary);
}

/* ─── Export PDF Link ──────────────────────────────────────── */
a[href*="export_pdf"] {
  display: inline-block;
  background: rgba(239,68,68,0.1);
  color: #f87171;
  border: 1px solid rgba(239,68,68,0.25);
  padding: 7px 16px;
  border-radius: var(--radius);
  text-decoration: none;
  font-size: 12px;
  font-weight: 600;
  cursor: pointer;
  transition: all 0.2s;
}

a[href*="export_pdf"]:hover {
  background: #ef4444;
  color: #fff;
  border-color: #ef4444;
}

.export-btn {
  display: inline-block;
  background: rgba(239,68,68,0.1);
  color: #f87171;
  border: 1px solid rgba(239,68,68,0.25);
  padding: 7px 16px;
  border-radius: var(--radius);
```

```
  font-size: 12px;
  font-weight: 600;
  cursor: pointer;
  transition: all 0.2s;
}

.export-btn:hover {
  background: #ef4444;
  color: #fff;
}

/* ─── Responsive ──────────────────────────────────────────────── */
@media (max-width: 768px) {
  .header { flex-direction: column; gap: 16px; text-align: center; }
  .user-info { text-align: center; }
  .stats { grid-template-columns: repeat(2, 1fr); }
  .filter-group { flex-direction: column; }
  .input-group { flex-direction: column; }
  .modal-content { padding: 20px; }
}

@media (max-width: 480px) {
  .stats { grid-template-columns: 1fr; }
  .container { padding: 14px 10px; }
  .section { padding: 16px; }
}
```

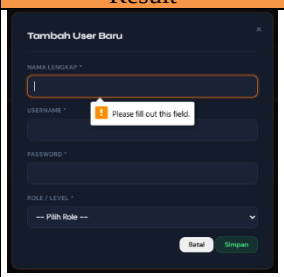
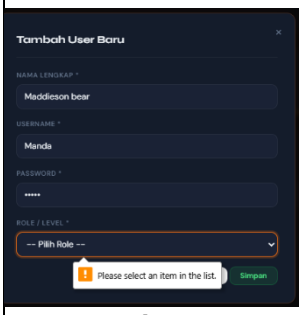
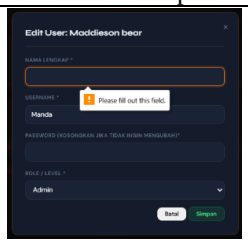
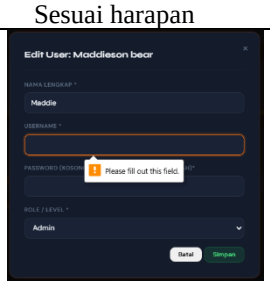
13. UJI COBA (BLACK BOX)

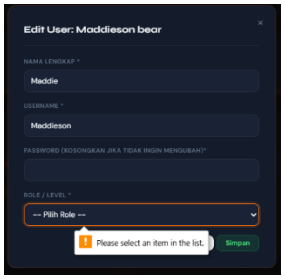
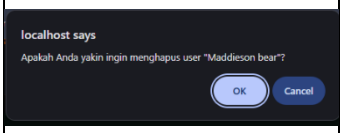
A. Pengujian form login

	Module	-condition	Test Case	st Scenario	Expected Rules	Actual Result
1.	Login	un sudah terdaftar	ak memasukkan username dan password	Akses halaman web lik tombol login	tem menampilkan peringatan “harap isi bidang ini”	 Sesuai harapan
2.	Login	Akun sudah terdaftar	ak memasukkan username	Akses halaman web 2.Isi password 3.Klik tombol login	tem menampilkan peringatan “harap isi bidang ini”	 Sesuai harapan

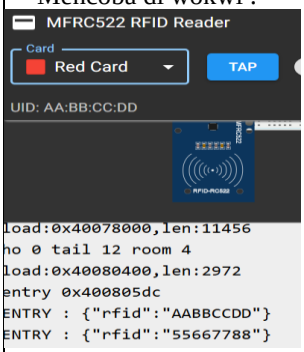
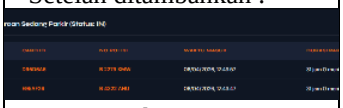
3.	Login	un sudah terdaftar	ak memasukkan password	Akses halaman web si username lik tombol login	tem menampilkan peringatan “harap isi bidang ini”	 Sesuai harapan

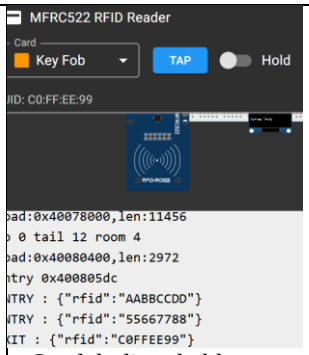
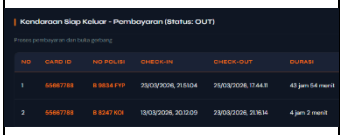
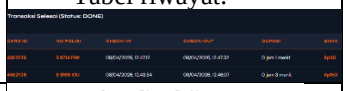
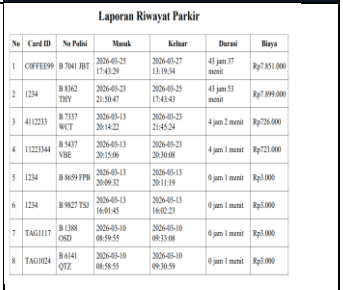
B. Pengujian From Tambah User dan Ubah User

.	Module	-condition	Test Case	st Scenario	Expected Rules	Actual Result
1.	Tambah user(admin)	er masuk sebagai Admin	nua field dikosongkan	nenekan tombol “Tambah user baru” lik “simpan”	tem menampilkan peringatan “harap isi bidang ini” la bagian nama lengkap	 Sesuai harapan
2.	Tambah user(admin)	User masuk sebagai Admin lalu menekan tombol “Tambah user baru”	gian opsi memilih role, tidak dipilih	si semua kolom kecuali opsi pilih role 2.Klik tombol “simpan”	tem menampilkan peringatan “pilih item pada daftar” la bagian role / level	 Sesuai harapan
3.	Edit user(Admin)	er masuk sebagai Admin lalu menekan tombol “edit”	ngosongkan bagian Nama lengkap	lik “simpan”	tem menampilkan peringatan “harap isi bidang ini”	 Sesuai harapan
4.	Edit user(Admin)	er masuk sebagai Admin lalu menekan tombol “edit”	ngosongkan bagian Username	lik “simpan’	tem menampilkan peringatan “harap isi bidang ini”	 Sesuai harapan

5.	Edit user(Admin)	er masuk sebagai Admin lalu menekan tombol “edit”	gian opsi memilih role, diubah menjadi opsi pilih role (role tidak dipilih)	si semua kolom kecuali opsi pilih role lik tombol “simpan	tem menampilkan peringatan “pilih item pada daftar”	 Sesuai harapan
6.	Edit user(Admin)	er masuk sebagai Admin lalu menekan tombol “edit” . ta sebelumnya sudah ditambahkan	ah data lama ke data yang baru	bah data lik tombol “simpan”	tem mengubah data, sesuai dengan data yang diubah oleh Admin	Sebelum diubah Setelah diubah Sesuai harapan
7.	Hapus User(Admin)	er masuk sebagai Admin lalu menekan tombol “Hapus”	nghapus salah satu user	lik tombol “simpan	tem menampilkan notifikasi konfirmasi	 Sesuai harapan

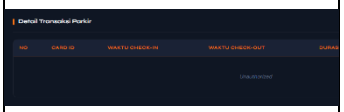
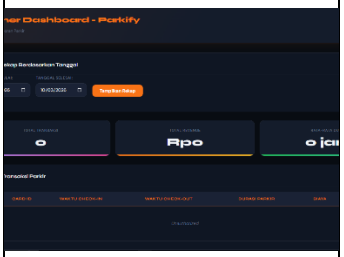
C. Pengujian Halaman Petugas

	Module	Pre-condition	Test Case	st Scenario	Expected Rules	Actual Result
1.	Checkin	er masuk sebagai Petugas	masukkan card id kedalam tabel checkin	ap kartu entry di wokwi ihat pada bagian tabel checkin	tem menampilkan data card id yang sudah di tap dari wokwi ke tabel checkin	Mencoba di wokwi :  Setelah ditambahkan :  Sesuai harapan
2.	Checkout	User masuk sebagai Petugas	mbali memasukkan card id yang sebelumnya sudah checkin ke	ap kartu exit di wokwi 2.Lihat pada bagian	tem menampilkan data card id yang sudah di tap dari wokwi ke	Mencoba di wokwi :

			tabel checkout	tabel checkout	tabel checkout	 Setelah ditambahkan :  Sesuai harapan
3.	Checkout	er masuk sebagai Petugas	ugas mengisi nominal pembayaran parkir dan menekan tombol "Buka palang & cetak struk"	Akses halaman web si username lik tombol login	tem Mencetak struk lalu memindahkan data checkout ke dalam tabel riwayat	Mengisi nominal : Cetak struk : SMART PARKING ard ID : C0FFEE99 o Pol : B 7041 JBT asuk : 25-03-2026 17:43 eluar : 27-03-2026 13:19 urasi : 43 jam 37 menit iaya : Rp7.851.000 HANK YOU FOR PARKING HERE HELL YEAH! Tabel riwayat: 
4.	Riwayat transaksi export data	er masuk sebagai Petugas	ngekspor data	lik tombol "Export data"	ngekspor data laporan yang ada di tabel riwayat	 Laporan Riwayat Parkir No | Card ID | No Polisi | Masuk | Keluar | Durasi | Biaya 1 | C0FFEE99 | B 7041 JBT | 2026-03-25 17:43:29 | 2026-03-27 13:19:34 | 43 jam 37 menit | Rp7.851.000 2 | 1234 | B 8362 THY | 2026-03-23 21:50:47 | 2026-03-25 17:43:43 | 43 jam 53 menit | Rp7.899.000 3 | 4112233 | B 7337 WCT | 2026-03-13 20:14:22 | 2026-03-23 21:45:24 | 4 jam 2 menit | Rp726.000 4 | 11223344 | B 5437 VBE | 2026-03-13 20:15:06 | 2026-03-23 20:30:08 | 4 jam 1 menit | Rp723.000 5 | 1234 | B 8659 FPB | 2026-03-13 20:09:32 | 2026-03-13 20:11:19 | 0 jam 1 menit | Rp3.000 6 | 1234 | B 9827 TSJ | 2026-03-13 16:01:45 | 2026-03-13 16:02:23 | 0 jam 1 menit | Rp3.000 7 | TAG1117 | B 1388 OSD | 2026-03-10 08:59:55 | 2026-03-10 09:33:08 | 0 jam 1 menit | Rp3.000 8 | TAG1024 | B 6141 QTZ | 2026-03-10 08:58:55 | 2026-03-10 09:30:59 | 0 jam 1 menit | Rp3.000 Sesuai harapan

D. Pengujian Filter Tanggal Owner

	Module	-condition	Test Case	st Scenario	Expected Rules	Actual Result
1.	ter rekap berdasarkan tanggal	er masuk sebagai owner	masukkan tanggal rekap	emasukkan tanggal mulai emasukkan tanggal selesai enekan tombol "Tampilkan rekap"	tem menampilkan data tanggal yang ada	Filter tanggal :

						Hasil :  Tidak Sesuai harapan
2.	ter rekap berdasarkan tanggal	User masuk sebagai owner	masukkan tanggal yang belum memiliki transaksi	Memasukkan tanggal mulai Memasukkan tanggal selesai 3.Menekan tombol "Tampilkan rekap"	tem menampilkan pesan pada tabel data detail transaksi parkir "Tidak ada transaksi pada tanggal yang dipilih" dan tidak menampilkan data apapun	 Sesuai harapan

14. REVISI

15. UPLOAD WEB KE GITHUB

https://github.com/Diraalthaf/Parkify.git

16. PEMBUATAN GUIDANCE

User Manual Book dari website ini berada di Github tempat dimana *source code* website Parkify – Smart Parking